Convention 1: Numerator and Denominator of Fractions

In writing a fraction, the numerator/top is always in a hidden set of parentheses, even if unseen. In writing a fraction, the denominator/bottom is always in a hidden set of parentheses, even if unseen.

Example 2. *The fraction* $\frac{2}{3}$ *can (and should!) be thought of as* $\frac{(2)}{(3)}$ *or* $(2)/(3)$

Example 3. *The fraction* $\frac{2}{3+x}$ *can (and should!) be thought of as* $\frac{(2)}{(3+x)}$ *or* $(2)/(3+x)$

Warning 4: Numerator and Denominator of Fraction

Be careful when rewriting fractions, in light of Convention 1

Example 5. *It is correct to think of* $\frac{2}{3+x}$ *as* $(2)/(3+x)$ *or even to drop the parentheses around the* 2 *and have*

$$2/(3+x)$$

but it is **incorrect** *to drop further parentheses, since*

$$2/3+x$$

is a different value, because the order of operations applied to $2/3+x$ *does division first and addition last, while the order of operations applied to* $2/(3+x)$ *does addition first and division last.*

Say someone owes you \$20. Suppose you owe one person \$1 and owe another person \$4. When all the money ends up where it should be, how much do you [really] have?

Example 6. *Suppose* $a = 20$ *and* $b = 1 + 4$.

1. *What is the value of* $a - b$*?*

2. *What is the value of* $20 - 1 + 4$*?*

3. *What is the point of this question?*

Habit 7

When subtracting several terms, ensure that the contents being subtracted is protected in a set of parentheses.

Percentages: discount

Example 8. *If a shirt has a price of* \$21.50 *but a* 20% *discount is being offered, how much money do you save off the original price?*

Example 9. *If a shirt has a price of* \$21.50 *but a* 20% *discount is being offered, what is the amount you actually pay for the shirt?*

Example 10. *If a toaster has a price of* \$8.40 *but a* 8% *discount is being offered, how much money do you save off the original price?*

Example 11. *If a toaster has a price of* \$8.40 *but a* 8% *discount is being offered, what is the amount you actually pay for the toaster?*

Example 12. *If a hat has a price of* h *but a* 6% *discount is being offered, how much money do you save off the original price?*

Example* 13. *If a hat has a price of* h *but a* 6% *discount is being offered, what is the amount you actually pay for the hat?*

Percentages: Tax

Example 14. *A laptop has a price of* \$1400 *and* 4% *sales tax will be added. How much money do you pay in sales tax?*

Example 15. *A laptop has a price of* \$1400 *and* 4% *sales tax will be added. How much total money do you pay, including sales tax?*

Example 16. *A laptop has a price of* L *and* 4% *sales tax will be added. How much money do you pay in sales tax?*

Example* 17. *A laptop has a price of* L *and* 4% *sales tax will be added. How much total money do you pay, including sales tax?*

Question 18

Why was the operation different for taxes than for discounts?

Percentages: tax and tip

Example 19. *A restaurant meal has a price of* \$15.80. *If* 5% *tax must be paid, how much money is paid in tax?*

Example 20. *A restaurant meal has a price of* \$15.80. *If* 5% *tax must be paid, how much total money is paid between the meal and tax?*

Example 21. *A restaurant meal has a price of* \$15.80. *If* 5% *tax must be* *paid, and* 15% *tip will be included on the post-tax amount. How much total money is paid in tip?*

Example 22. *A restaurant meal has a price of* \$15.80*. If* 5% *tax must be paid, and* 15% *tip will be included on the post-tax amount. How much total money is paid in meal, tax, and tip?*

Example 23. *A restaurant meal has a price of M. If* 5% *tax must be paid, how much money is paid in tax?*

Example 24. *A restaurant meal has a price of M. If* 5% *tax must be paid, how much total money is paid between the meal and tax?*

Example 25. *A restaurant meal has a price of M. If* 5% *tax must be paid, and* 15% *tip will be included on the post-tax amount. How much total money is paid in tip?*

Example 26. *A restaurant meal has a price of M. If* 5% *tax must be paid, and* 15% *tip will be included on the post-tax amount. How much total money is paid in meal, tax, and tip?*

Habit 27: Putting expressions into a calculator

Just as with Habit **??**, always write expressions using order of operations when inputting mathematics into a calculator. This may require parentheses so that what you intend to say follows Convention **??**.

Example 28. *Consider the expression* $\frac{2}{3x}$.

- *How should this be entered on a calculator?*
- *What is a dangerous way to write this expression on paper, and why is it dangerous?*

Example 29. *State the factors of* $6xy$

Example 30. *State the factors of* $3x^2$

Example* 31. *What are the common factors of* $3x^2$ *and* $6xy$*? What is the greatest common factor of* $3x^2$ *and* $6xy$*?*

Concept 32: Factoring produces factors

The process of factoring results in having factors (things that are multiplied together).

Example 33. *Factor* 36

There are two ways to compute the entire area. First, the large rectangle has one side length of a, and due to the fact that the picture form of adding is "gluing sticks together" the other side length of the big rectangle is $b + c$. So, the area of the big rectangle is the product of these side lengths, namely $a(b + c)$. The other way to compute the big area is to add the areas of the two smaller rectangles: the rectangle on the left has area ab and the rectangle on the right has area ac. So, the combined area is $ab + ac$. Setting equal the two different ways we computed the same area, $a(b + c) = ab + ac$.

Example 34. *Factor* $3x^2 + 6xy$

Answer: We saw in Example 31 that $3x$ is a common factor of $3x^2$ and $6xy$. So $3x^2 + 6xy = 3x(x + 2y)$

Give a geometric explanation why
$(a + b)(c + d) = ac + ad + bc + bd$ is true.

The way we presented FOIL (**F**irst, **O**uter, **I**nner, **L**ast), we need all plus signs, which doesn't always happen. In that case, we think about distributing instead. For example, suppose we want to multiply $(2x + 4y) \cdot (5z - 6x)$. We have

$$\begin{aligned}
(2x + 4y)\cdot(5z + (-6x)) &= (2x + 4y)\cdot 5z + (2x + 4y)\cdot(-6x) \\
&= 5z(2x + 4y) + (-6x)(2x + 4y) \\
&= (5z)\cdot(2x + 4y) + (-6x)\cdot(2x + 4y) \\
&= (5z)\cdot 2x + 5z \cdot 4y + (-6x)\cdot 2x + (-6x)\cdot(4y) \\
&= 10xz + 20zy - 12x^2 - 24xy
\end{aligned}$$

Factoring

Factoring is the opposite of distribution. We have the following properties:

Formula 35

$x^2 + 2xy + y^2 = (x + y)(x + y)$

Formula 36

$x^2 - 2xy + y^2 = (x - y)(x - y)$

Formula 37: Difference of Squares

$x^2 - y^2 = (x + y)(x - y)$

Notice that going left to right we are factoring, but going right to left we are distributing.

Warning 38

$(x + y)^2 \neq x^2 + y^2$

Instead, $(x + y)^2 = x^2 + 2xy + y^2$.

Example 39. *Factor* $x^2 + 10x + 21$

Example 40. *Factor* $9x^2 - 36$

Example 41. *Factor* $4x^2 - 12x + 9$

Example 42. *Factor* $x^3 + 5x^2 - 14x$

Example 43. *Factor* $4x^3 - 12x^2 + 5x - 15$

Example 44. *SAVE this for AFTER quadratic factoring Simplify* $\frac{x-5}{x^2-25}$

Example 45. *Solve* $\frac{3}{x+1} - \frac{1}{2} = \frac{1}{3x+3}$

When multiplying two fractions, just multiply straight across, just as we saw in Formula **??**. We still do this, even when there are variables in fractions:

$$\begin{aligned}
\frac{(x+2)}{(2x-3)} \cdot \frac{(2x^2-3x)}{(x^2-4)} &= \frac{(x+2)(2x^2-3x)}{(2x-3)(x^2-4)} \\
&= \frac{(x+2)x(2x-3)}{(2x-3)(x-2)(x+2)} \\
&= \frac{\cancel{(x+2)}x\cancel{(2x-3)}}{\cancel{(2x-3)}(x-2)\cancel{(x+2)}} \\
&= \frac{x}{x-2}
\end{aligned}$$

When dividing two fraction expressions, “flip and multiply”, as we saw in Formula **??**:

$$
\begin{aligned}
\frac{4x^2-9}{2x^2+7x+6} \div \frac{27x^4+8x}{9x^4-6x^3+4x^2} &= \frac{4x^2-9}{2x^2+7x+6} \cdot \frac{9x^4-6x^3+4x^2}{27x^4+8x} \\
&= \frac{(4x^2-9)(9x^4-6x^3+4x^2)}{(2x^2+7x+6)(27x^4+8x)} \\
&= \frac{(2x-3)(2x+3)x^2(9x^2-6x+4)}{(2x^2+7x+6)x(27x^3+8)} \\
&= \frac{(2x-3)(2x+3)x^2(9x^2-6x+4)}{(2x+3)(x+2)x(3x+2)(9x^2-6x+4)} \\
&= \frac{(2x-3)\cancel{(2x+3)}x^{\cancelto{1}{2}}\cancel{(9x^2-6x+4)}}{\cancel{(2x+3)}(x+2)\cancel{x}(3x+2)\cancel{(9x^2-6x+4)}} \\
&= \frac{(2x-3)x}{(x+2)(3x+2)}
\end{aligned}
$$

When adding or subtracting two fractional expressions, remember to get a common denominator, as we saw in Formula **??**:

$$
\begin{aligned}
\frac{t}{t+3} + \frac{4t}{t-3} - \frac{18}{t^2-9} &= \frac{t}{t+3} \cdot \frac{t-3}{t-3} + \frac{4t}{t-3} \cdot \frac{t+3}{t+3} - \frac{18}{t^2-9} \\
&= \frac{t(t-3)}{t^2-9} + \frac{4t(t+3)}{t^2-9} - \frac{18}{t^2-9} \\
&= \frac{t(t-3)+4t(t+3)-18}{t^2-9} \\
&= \frac{t^2-3t+4t^2+12t-18}{t^2-9} \\
&= \frac{5t^2+9t-18}{t^2-9}
\end{aligned}
$$

Example 46. *Simplify* $\dfrac{4}{(5s-2)^2} + \dfrac{s}{5s-2}$.

Example 47. *Simplify* $\dfrac{4x+12}{x^2+6x+9} + \dfrac{5x}{x^2-9} + \dfrac{7}{x-3}$.

Example 48. *Simplify* $\dfrac{\frac{r}{s}+\frac{s}{r}}{\frac{r^2}{s^2}-\frac{s^2}{r^2}}$.

Example 49. *Simplify* $\dfrac{\frac{1}{x+h}-\frac{1}{x}}{h}$.

Solving equations, continued

Example 50. *Find values for x and y that satisfy both $2x+y = 1$ and $3x+4y = 14$*

Example 51. *Find values for x and y that satisfy both $2x - y = 5$ and $x + 4y = 7$*

Example 52. *Solve*[1] *for T in $(-9.35)(1000) = (335)(0.248)(T - 391)$.*

[1] Finding T is the final temperature in a heat loss problem from **chemistry**.

Example 53. *Solve*[2] *for x in $0.500 - 2x = 0.215$*

[2] This equation came from a chemistry class.

Example 54. *You have 1600 dollars to buy a laptop, but have to also pay a 4% sales tax. What is the pre-tax price of a laptop that you can afford?*

Recall Example 17

Example 55. *You have 10 dollars to buy a hat. Hats are being offered at a 6% discount off original prices. What is the maximum original price of a hat we can afford?*

Recall Example 13

Suggested practice problems:

- On Monday, you go to Culver's. The Mushroom & Swiss Burger Value Basket is listed as costing $8.75, but that is the pre-tax amount. If there is a 6% sales tax, then how much total needs to be paid?
- On Tuesday, you return to Culver's. Suppose you have exactly $15 in your wallet. If there is a 6% sales tax, then what is the maximum [pre-tax] amount of food you can afford to pay in cash?
- During a trip to Oregon, you go to a department store and find a shirt you really like. The shirt has a $15 price tag on it, but there is a 6% off sale. What is the final price you end up needing to pay for the shirt? (Oregon has no sales tax.)
- During a trip to Montana, you go to a department store and have decided in advance that you will spend no more than $130. Luckily, everything on the store is part of a 20% off sale. What is the maximum amount that you can have (as the amounts of all price tags added together) to stay within your $130 spending limit? (Montana has no sales tax.)
- Sydney is going to a restaurant and they have 30 dollars with them. If Sydney plans to tip 18%, then what is the largest bill (prior to calculating the tip) that they can afford? (Assume that there is no sales tax.)

- A couple does not wish to spend more than \$50 for dinner at a restaurant. If a sales tax of 8% is added to the bill, and they plan to tip 15% after the tax has been added, what is the most they can spend for the meal?
- A worker's weekly take-home pay is 120 dollars from a part time job, after deductions totaling 30% of the gross pay have been subtracted. What is the weekly gross pay?

Formula 56: $d = rt$

(distance) = (rate) × (time)

Warning 57

If there are two journeys,

- do not assume the two journeys have the same distance
- do not assume the two journeys have the same rate
- do not assume the two journeys have the same time

Example 58. *Two cyclists,* 63 *miles apart, start riding toward each other at the same time. One cycles twice as fast as the other. If they meet* 2 *hours later, at what average speed is each cyclist traveling?*

Example 59. *Wendy took a trip whose total distance was* 80 *miles. She traveled part of the way by bus, which arrived at the train station just in time for Wendy to complete her journey by train. The bus averaged* 25 *miles per hour, and the train averaged* 60 *miles per hour. The entire trip took* 1.5 *hours. How long did Wendy spend on the train?*

Example 60. *Google says the distance between La Crosse and Milwaukee is 210 miles. How much faster do you get to Milwaukee from here by driving 75 mph instead of the speed limit? (The speed limit on I-90 is 70 mph). Out of curiosity, what is the final answer in minutes?*

Example 61. *I have a beaker (Beaker 1) with* 8 *with liters of fluid. If* 50% *of the fluid in the beaker is acid, how many liters of acid are in the beaker?*

Formula 62: Concentration equation

$$(\text{amount solution})(\text{concentration}) = (\text{amount acid})$$

This concentration equation can be used on every container.

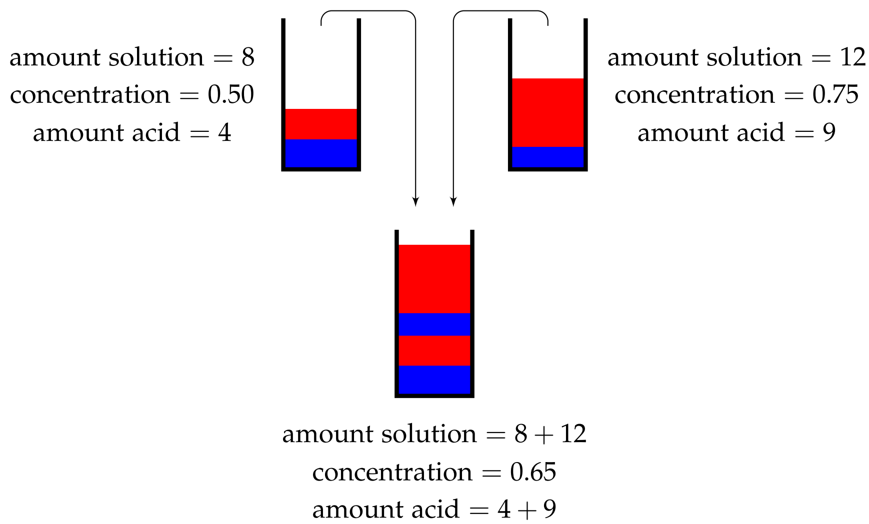
Example 63. *Beaker 2 contains* 12 *liters of liquid. If* 75% *of the liquid is acid, how many liters of acid are in Beaker 2?*

Example 64. *Beaker 1 contains* 8 *with liters of liquid, and* 50% *of the liquid in the beaker is acid. Beaker 2 contains* 12 *liters of liquid, and* 75% *of the liquid is acid.*

If the contents of Beaker 1 and Beaker 2 are dumped into an empty bucket, what percentage of the bucket is acid?

Warning 65

The percentages of concentration cannot be added.



Example 66. *How many liters of a* 20% *acid solution should be mixed with* 80 *liters of a* 61% *acid solution to make a* 45% *acid solution?*

Example 67. *How many liters of a* 20% *acid solution and how many liters of a* 61% *acid solution should be mixed together to make* 80 *liters of a* 45% *acid solution?*

Method 68: Mixture

Every mixture problem can be rephrased as taking the contents of one beaker and the contents of another beaker and dumping both into a bucket. Since the concentration equation must be applied to each container, for each container, you must keep track of the amount of solution, the concentration of acid, and the amount of acid. Be sure to address the following:

- Apply the concentration equation to Beaker 1.
- Apply the concentration equation to Beaker 2.
- Apply the concentration equation to the final bucket.
- The amount of solution in the bucket is the sum of the amount of solution in Beaker 1 and the amount of solution in Beaker 2.
- The amount of acid in the bucket is the sum of the amount of acid in Beaker 1 and the amount of acid in Beaker 2.
- You will always obtain two expressions for the amount of acid in the bucket, so equate these.

Extra practice problems:

1. A jar has liquid. Some of the liquid is acid, and some of the liquid is non-acid. If the jar has 8 liters of liquid and 50% of the jar is acid, how many liters of acid are in the jar?
2. If a 5 liter jar is 84% acid, how much of the jar (in liters) is acid?
3. You have an empty bucket. An 8 liter jar which is 50% acid is dumped into the bucket. A 5 liter jar which is 84% acid is also dumped into the bucket.
   (a) What is the total volume of liquid in the bucket (in liters)?
   (b) How many liters of acid are in the bucket?
   (c) What is the concentration (as a percent) of acid in the bucket?
4. What amount of a 60% acid solution must be mixed with a 30% solution to produce 24 liters of a 50% solution?
5. A manufacturer of soft drinks advertises their orange soda as “naturally flavored,” although it contains only 5% orange juice. A new federal regulation stipulates that to be called “natural,” a drink must contain at least 10% fruit juice. How much pure orange juice must this manufacturer add to 500 gallons of orange soda to conform to the new regulation?

Additional practice problems:

Example 69. *A woman earns 15% more than her husband. Together they make $69,875. What is the husband's salary?*

Example 70. *A plumber and his assistant work together. The plumber charges $45 an hour and his assistant charges $25 an hour. The plumber works twice as long as his assistant on a job and the final bill for both workers is$4,025. How long did the plumber and his assistant work?*

Example 71. *A man walking away form a lamppost with a light source 6 meters above the ground. The man is 2 meters tall. How long is the man's shadow when he is 10 meters from the lamppost?*

Example 72. *A pot contains 6 liters f brine at a concentration of 120 g/L. How much water should be boiled off to increase the concentration to 200 g/L?*

Example 73. *Two cyclists, 90 miles apart, start riding toward each other at the same time. One cycles twice as fast as the other. If they 2 hours later, at what average speed is each cyclist traveling?*

Example 74. *If Ben invests $4000 at 4% interest per year, how much additional money must he invest at 5.5% annual interest to ensure that the interest he receives each year is 4.5% of the total amount invested?*

Exponents

Example 75. *Given* $P = 1000$ *and* $r = 0.04$ *and* $t = 5$, *simplify*[3] *the value of* Pe^{rt}.

[3] This is a **finance** problem, and the value answers the following question: if $1000 is deposited now in an account earning 4% interest compounded continuously, what will be account balance in 5 years?

Formula 76

Let a, b, x, y be real numbers. We have the following properties:

- $(a^x)^y = a^{xy}$
- $(a^x)(a^y) = a^{x+y}$
- $a^0 = 1$ (except possibly if $a = 0$)
- $a^{-n} = \dfrac{1}{a^n}$
- $(a \cdot b)^n = a^n \cdot b^n$
- $\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$
- $a^n = (a^{-n})^{-1} = \dfrac{1}{a^{-n}}$

Example 77. *Simplify* $(a^3)^5$

Example 78. *Simplify* a^3a^5

Example 79. *Simplify* $a^{2/3}a^{1/5}$

Example 80. *Simplify* $(xy^4)^3x^8(x^3z)^{10}$

Example 81. *Simplify* $\dfrac{c^7}{c^{\frac{1}{2}}}$

Example 82. *Simplify* $\dfrac{c^{14}}{c^{-2}}$

Radicals

Concept 83: What does a radical mean?

$\sqrt[n]{a}$ asks for the value that fills in the blank:

$$(\qquad\qquad)^n = a$$

If two numbers can fill in the blank (positive and negative), then by definition, the positive number is the answer.

Example 84. *Find* $\sqrt[2]{64}$

Example 85. *Find* $\sqrt[3]{64}$

Concept 86

$\sqrt{a} = \sqrt[2]{a}$

If the small number outside of the radical is missing, it is a hidden 2.

Language Discussion 87

$\sqrt[n]{a}$ should be said "nth root of a

Language Discussion 88

$\sqrt[2]{a}$ and $\sqrt{a}$ can be said either "2nd root of a" or "square root of a

Language Discussion 89

$\sqrt[3]{a}$ is spoken "3rd root of a" but cannot be called "3 square root of a"

Language Discussion 90

$\sqrt[7]{a}$ is spoken "7th root of a" but cannot be called "7 square root of a"

Formula 91

$\sqrt[n]{ab} = \sqrt[n]{a}\sqrt[n]{b}$

Formula 92

$\sqrt[n]{a^n} = a$

Example 93. *Simplify* $\sqrt{18}$

Example 94. *Simplify* $\sqrt{20x^9}$

Example 95. *Simplify* $\sqrt[3]{80a^9b^{10}c^{11}d^{12}}$

Radicals (square roots $\sqrt{\ }$, cube roots $\sqrt[3]{\ }$, etc) are just ways to represent certain exponents. If you see a radical, turn it into its corresponding exponent. This will help you most of the time.

Formula 96

$a^{1/n} = \sqrt[n]{a}$ (when it this makes sense)

When working with radicals, we need to remember that we cannot have a negative under an even root. However, you *can* have a negative under an odd root. The reason for this is the following: Suppose you have $\sqrt{-2} = x$. Then, this means that $(\sqrt{-2})^2 = x^2$. So, $-2 = x^2$. This is an issue because x^2 is always positive. On the other hand, suppose you have $\sqrt[3]{-27} = x$. Then, this means that $(\sqrt[3]{-27})^3 = x^3$. Thus, $-27 = x^3$. This demonstrates why we can have negatives with odd roots.

$x^{1/5} = \sqrt[5]{x}$

$$27^{-2/3} = \frac{1}{27^{2/3}} = \frac{1}{(27^{1/3})^2} = \frac{1}{(\sqrt[3]{27})^2} = \frac{1}{3^2} = \frac{1}{9}$$

Warning 97

$$x^{1/2} \neq \frac{1}{x^2}.$$

Example 98. *Write* $\dfrac{\sqrt[4]{x}}{\sqrt[3]{x}}$ *with only 1 radical.*

Example 99. *Simplify* $16^{-3/4}$.

Example 100. *Simplify* $(-3x^{-2})(4x^4)$.

Example 101. *Simplify* $\left(\dfrac{4a^2b}{a^3b^3}\right)\left(\dfrac{5a^2b}{2b^4}\right)$.

Example 102. *Simplify* $(-6x^{7/5})(2x^{8/5})$.

Example 103. *Simplify* $\left(\dfrac{-y^{3/2}}{y^{-1/3}}\right)^{-3}$.

Example 104. *Simplify* $\sqrt[5]{\dfrac{3x^{11}y^3}{9x^2}}$.

Example 105. *Simplify* $\sqrt{3a^2b^3}\sqrt{6a^5b}$

Example 106. *Simplify* $\sqrt[3]{320}$

Example 107. *Simplify* $\sqrt[3]{16x^3y^8z^4}$

Example 108. *Simplify* $(r^2s^6)^{1/3}$

Example 109. *Simplify* $\left(\dfrac{2x^{2/3}}{y^{1/2}}\right)^2\left(\dfrac{3x^{-5/6}}{y^{1/3}}\right)$

The Curse of Distribution

In general, $(x \pm y)^n \neq x^n \pm y^n$. For example,

$$(x+y)^2 \neq x^2 + y^2.$$

In reality,

$$(x+y)^2 = (x+y)(x+y) = x^2 + 2xy + y^2$$

This is a property of *EXPONENTS*. And, **exponents do <u>not</u> distribute over addition or subtraction**. So, since radicals are *really* exponents, radicals **DO NOT** distribute over addition and subtraction.

Warning 110

This means
$$\sqrt[n]{x \pm y} \neq \sqrt[n]{x} \pm \sqrt[n]{y}$$

Here are some examples:

- $\sqrt{1+x^2} = 1+x$ False
- $\sqrt{x^2+y^2} = x+y$ False
- $\sqrt{4+x} = 2+\sqrt{x}$ False
- $\sqrt{4 \cdot x} = 2 \cdot \sqrt{x}$ True
- $\sqrt{x^2y^2} = xy$ False

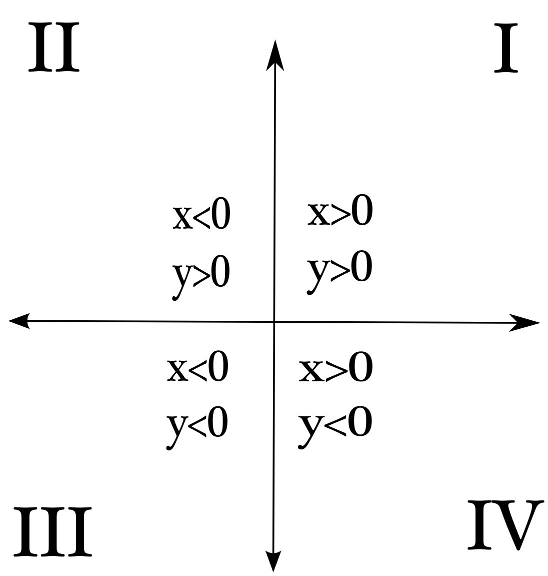
Formula 111

If $x \geq 0$, then $(\sqrt{x})^2 = \sqrt{x^2} = x$

The xy-plane

xy-plane and the Quadrants

The xy-plane can be divided into four quadrants which are described by the behavior of x and y.



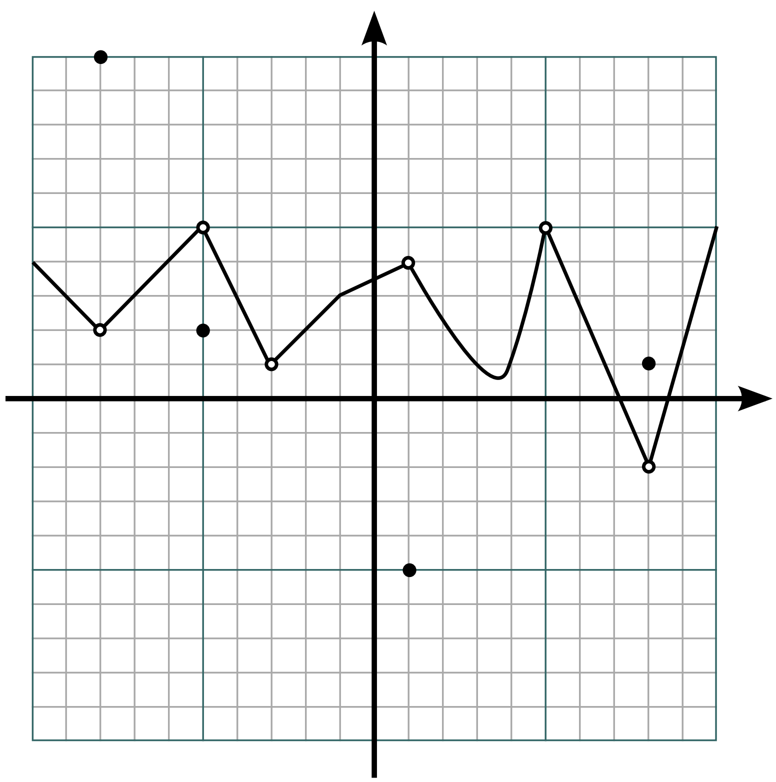
A point always has an x-coordinate and a y-coordinate, wrapped inside a set of parentheses.

Concept 112

The information about a point can be written by labelling coordinates by the point, or putting tick marks on the axes.

Example 113. *Provide both representations of the point* $(8, 3)$.

In the picture below, the point $(-5, 2)$ is part of the graph, but the point $(-5, 5)$ is not. The "unfilled dot" there is telling is we get all the points between $(-8, 2)$ and $(-5, 5)$ but not the point $(-8, 2)$ itself or the point $(-5, 5)$ itself. On the other hand, the graph includes the point $(-1, 3)$, even though the point hasn't been drawn in thick.



Formula 114: Distance Formula

iven two points (x_1, y_1) and (x_2, y_2) the **distance** between these points is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This comes from the Pythagorean Theorem which states that a right triangle with sides a and b and hypotenuse c satisfies the equation

$$a^2 + b^2 = c^2,$$

Example 115. *Show that $(-6, 3)$, $(-1, 5)$ and $(3, -5)$ are vertices of a right triangle. Then, find the area of that triangle.*

Example 116. *Find x so that the distance between the two points, $(x, 3)$ and $(4, 5)$ is 5.*

Formula 117: Midpoint Formula

Given two points (x_1, y_1) and (x_2, y_2) the **midpoint** between these points is given by

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

Note that this is a point.

Example 118. *Given points $A(5, -8)$ and $B(-6, 2)$, find the point on the line segment AB that is $\frac{3}{4}$ of the way from A to B.*

Example 119. *Find the point on the line $y = x$ that is equidistant from $(-2, 4)$ and $(4, 5)$.*

Graphs of equations

This section focuses on equations in x and y.

Example 120. $x^3 + xy + 6 = x + xy^2$ *is an equation in x and y*

Every equation in x and y can be graphed on the xy-plane.

Concept 121: How is an equation in x and y related to its graph?

The point (x, y) is part of the graph if and only if (x, y) satisfies the equation.

Example 122. *Is the point $(2, 3)$ part of the graph of $x^3 + xy + 6 = x + xy^2$?*

Example 123. *Is the point* $(2,1)$ *part of the graph of* $x^3 + xy + 6 = x + xy^2$*?*

Example 124. *Is the point* $(1,4)$ *part of the graph of* $x^2 + 2y = 20 - 4x - y^2$*?*

Example 125. *Is the point* $(1,3)$ *part of the graph of* $x^2 + 2y = 20 - 4x - y^2$*?*

https://www.desmos.com/calculator/b5c6wq2rtp

Example 126. *Find three points on the graph of* $x = y^2 - 5$

Example 127. *Find three points on the graph of* $2y = x^3$

Graphs of lines

Lines

In order to find an equation of a line, you need are 2 points or a point and a slope. If you are working on a problem and it asks you to find an equation of a line, you need to ask yourself if you already have 2 points or if you have a point and a slope. This will determine which strategy you should use to answer the problem.

Slopes of Lines

In order to find the slope of a line, you will need only two points, generically labeled (x_1, y_1) and (x_2, y_2). Once you have these two points, the slope is given by a formula:

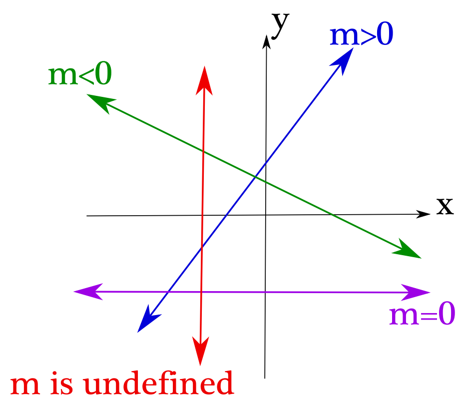
Definition 128: Slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x} = \frac{\triangle y}{\triangle x}$$

The phrase "rise over run" is meant to help so that we don't accidentally put x-values in the numerator. The word "rise" lets you know we are talking about the change in height, and the height is given by the y-value. Slope is "rise over run" not "run over rise"

Warning 129: Incorrect slope

$$m \neq \frac{x_2 - x_1}{y_2 - y_1}$$



Types of Equations

There are several types of equations of lines and ALL of them are okay to use. (However, sometimes directions to homework problems will specifically ask for one equation or another as an answer. Or, it could be that it makes sense to use one equation over another.)

Definition 130: Point-Slope form of a line

$$y - y_1 = m(x - x_1)$$

where m is the slope and (x_1, y_1) is *any* point on the line.

Example 131. $y - 2 = \frac{3}{4}(x - 5)$ *describes the line of slope* $\frac{3}{4}$ *going through the point* $(5, 2)$.

Definition 132: Slope-Intercept form of a line

$$y = mx + b$$

where m is the slope and b is the y-intercept

Example 133. $y = \frac{6}{7}x + 8$ *describes the line of slope* $\frac{6}{7}$ *with y-intercept* 8.

Warning 134

Note that m is the slope in $y = mx + b$, but mx is not the slope.

Example 135. *Earlier, we saw that* $y = \frac{6}{7}x + 8$ *describes the line of slope* $\frac{6}{7}$*, but it would be incorrect to report the slope as* $\frac{6}{7}x$*. In the format*

$y = mx + b$, it is the content **before** *the x that is the slope, but without including x itself.*

Example 136. *Find the line that passes through the points $(-2, 4)$ and $(9, 2)$.*

Example 137. *Find the line that passes through the point $(1, 3)$ with slope $-\frac{1}{4}$*

Example 138. *Find the line that passes through the point $(2, 5)$ with x-intercept 4.*

Example 139. *Find the slope of the line given by $3x + 4y = 9$.*

Example 140. *Does the line given by $3x + 4y = 9$ pass through the point $(1, 1)$? Explain your reasoning.*

Example 141. *Does the line given by $3x + 4y = 9$ pass through the point $(3, 0)$? Explain your reasoning.*

Example 142. *In finding the line that passes through the points $(-2, 4)$ and $(9, 2)$, does it matter which point is (x_1, y_1) and which point is (x_2, y_2)? Explain.*

Fact 143: Parallel lines

Two lines are parallel if they have the same slope.

https://www.desmos.com/calculator/taifhuenas

Fact 144: Perpendicular lines

Two lines are perpendicular if their slopes are negative reciprocals. In other words, if line 1 has slope m_1 and line 2 has slope m_2, then

$$m_1 = \frac{-1}{m_2} \quad \text{or} \quad m_1 \cdot m_2 = -1$$

https://www.desmos.com/calculator/i6svrb0s6u

Example 145. *Sketch a graph of the line that passes through the points $(-2, 4)$ and $(9, 2)$.*

Example 146. *Sketch a graph of the line line that passes through the point* $(1,3)$ *with slope* $\dfrac{-1}{4}$.

Example 147. *Sketch a graph of the line that passes through the point* $(2,5)$ *with x-intercept 4.*

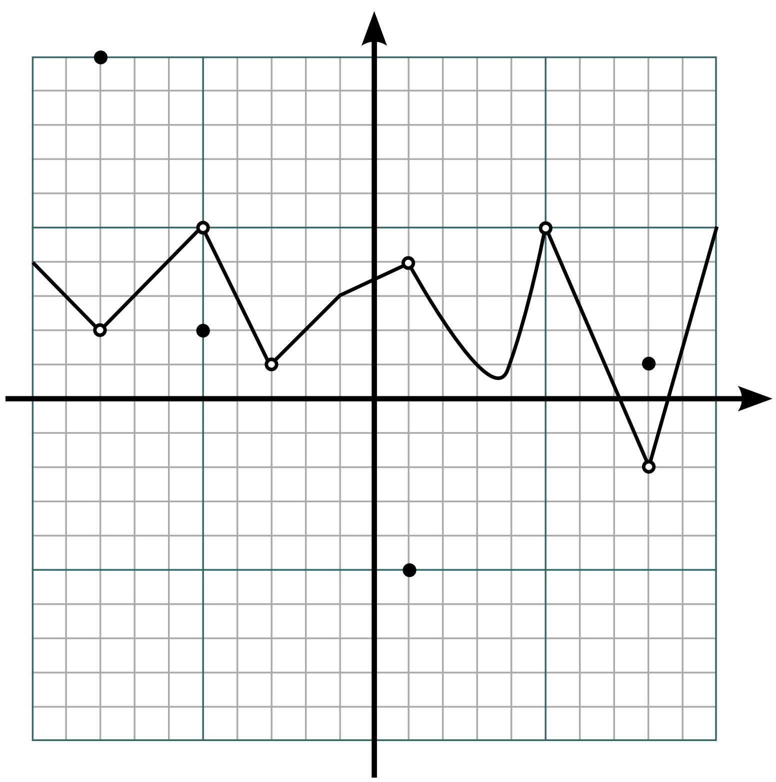
Example 148. *Find an equation of the line which is parallel to* $3x+2y=1$ *that passes through the point* $(1,1)$. *Then, sketch a graph of both lines.*

Example 149. *Find an equation of the line that is perpendicular to* $y = 3x+1$ *that passes through the point* $(-2,1)$. *Then, sketch a graph of both lines.*

Example 150. *Sketch the line given by* $y = 2x+1$.

Example 151. *Sketch the line given by* $y-3 = 2(x+1)$.

Example 152. *Sketch the line given by* $2x-3y=1$



Example 153. *In the graph above, find the y-value of the point in the graph when the x-value is 6.*

Applications of geometry in the xy-plane

Example 154. *The two lines* $y = \dfrac{3}{8}x+17$ *and* $y = -\dfrac{1}{4}x+42$ *intersect*[4] *at*

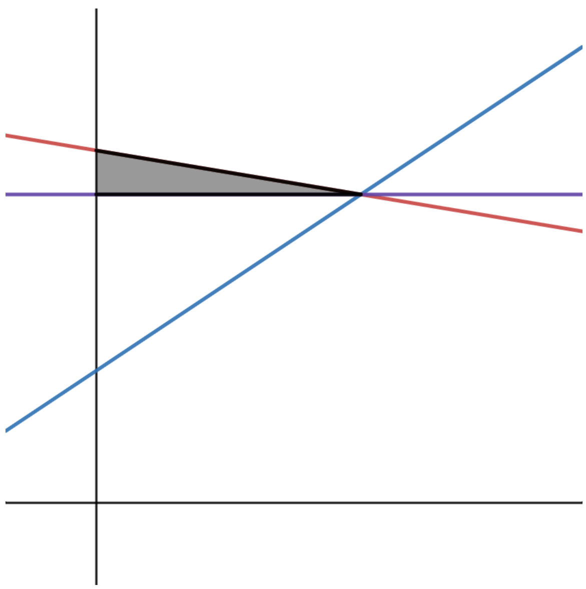
[4] This is a typical **business** problem in the area of supply and demand. In business, supply always has positive slope and demand has negative slope. So, if $y = \dfrac{3}{8}x+17$ represents supply and $y = -\dfrac{1}{4}x+42$ represents demand, then the intersection point that

a point. Find the point.

Example 155. *Find the area of the region below $y = 2x$, above the x-axis, and between the lines $x = 7$ and $x = 10$. (There are at least two ways to do this.)*

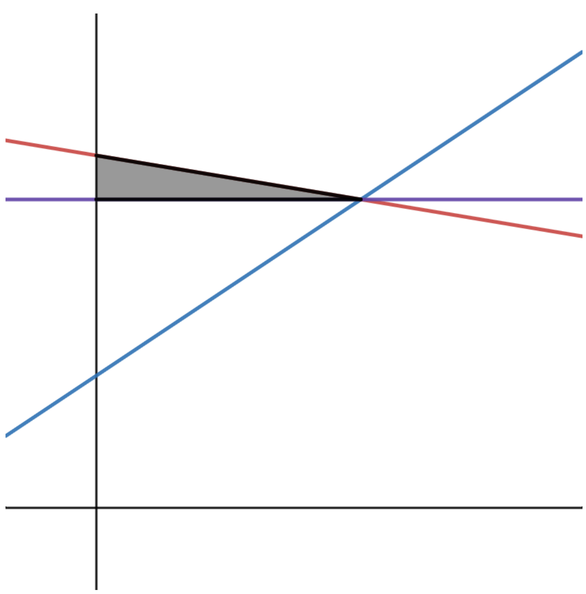
Example 156. *In the image below, the blue line has equation $y = \frac{2}{3}x + 3$ and the red line has equation $y = -\frac{1}{6}x + 8$. The purple line is a horizontal line drawn by the intersection of the red and blue lines. Find the area of the shaded region*[5].

[5] In **economics**, if the blue line represents supply and the red line represents demand, then the area requested represent **producer surplus**.



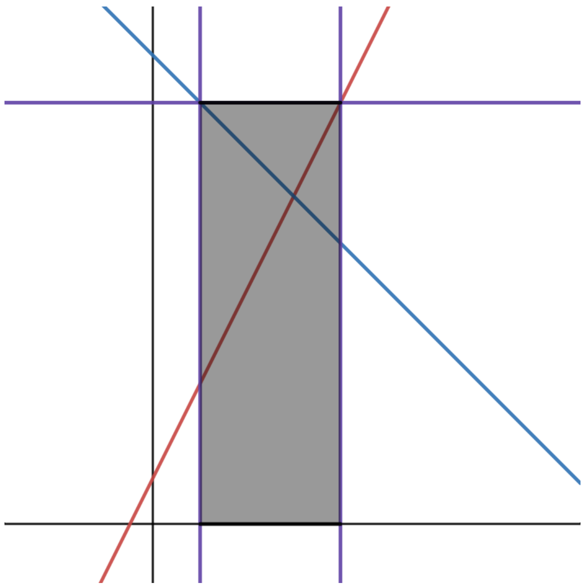
Example 157. *In the image below, the blue line has equation $y = mx + b$ and the red line has equation $y = nx + c$. Find the area*[6] *of the shaded region in terms of m, b, n, and c.*

[6] In **economics**, if the blue line represents supply and the red line represents demand, then the area requested represent **producer surplus**.



Example 158. *In the image below, the blue line has equation $y = 10 - x$ and the red line has equation $y = 2x + 1$. The purple horizontal line is $y = 9$, and the vertical purple lines are created by the intersection of the horizontal line with the red and blue lines. Find the area of the shaded region. The purple line is a horizontal line drawn by the intersection of the red and blue lines. Find the area[7] of the shaded region.*

[7] In **economics**, this area represents **government expenditures**.

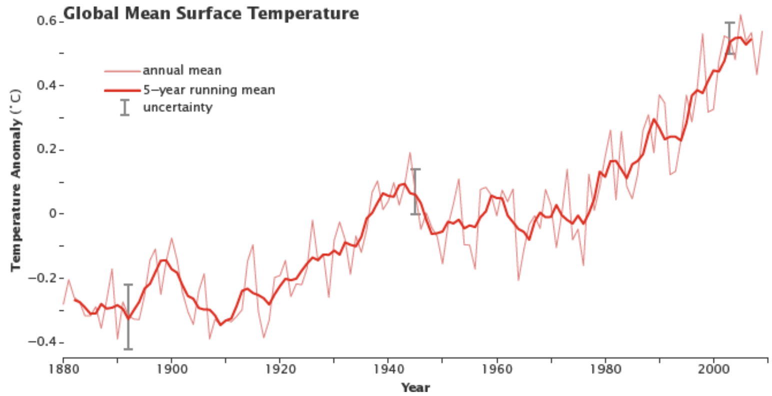


Graph interpretation

An all areas where math gets used (economics, biology, chemistry, physics, sociology, weather forecasting, etc.) graphs are presented to show how one quantity relates to another. One quantity is labeled on the x-axis, and one on the y-axis.

Examine the following graph from NASA, originally found at

`earthobservatory.nasa.gov/features/GlobalWarming/page2.php`

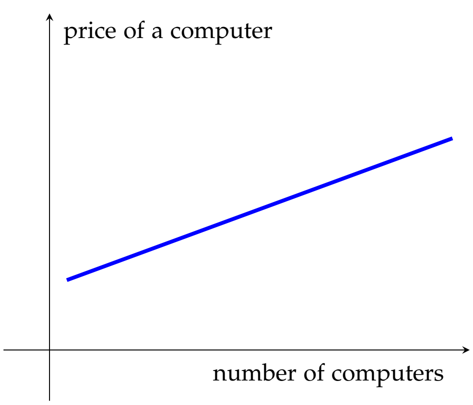


The picture above has the complication that two graphs are presented together. Let's look at the thicker graph. What we are seeing is that the further into the future we look (the year, which is the x-value, increases), it gets hotter. (I say "hotter" because the y-axis represents temperature, and bigger temperature values means hotter rather than colder.)

Because people talk about global warming, it is too easy to come up with the correct thing to say: it is easy to say what the graph is saying without actually looking at the graph. However, in future classes where you will look at an interpret graphs, you will probably not always get to rely on your intuition.

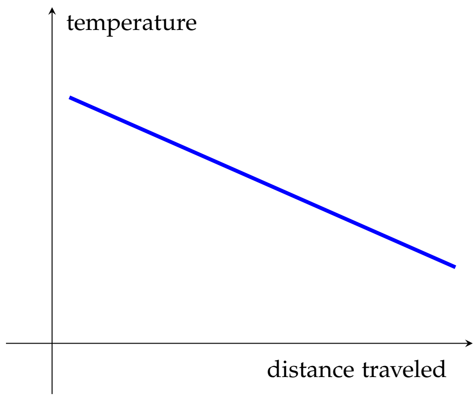
Here is an actual example from **economics**, slightly simplified.

Example 159. *Interpret the graph below*



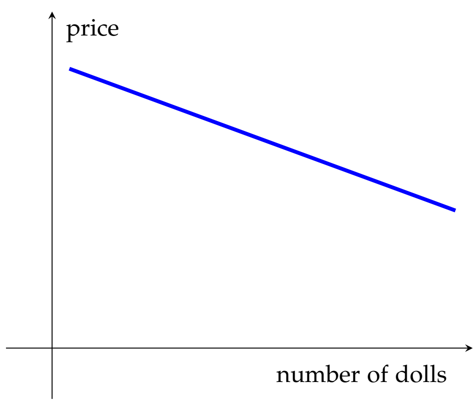
Answer: If there are more computers, then each computer gets

Example 160. *Interpret the graph below*



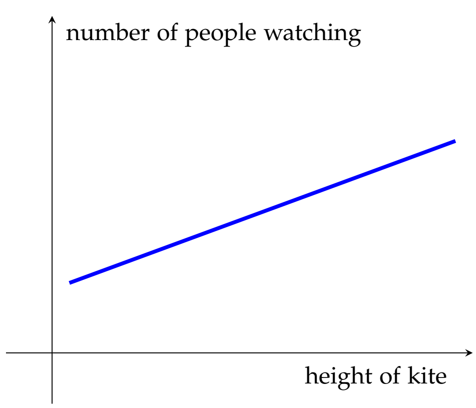
The longer the distance traveled, the _______________ it gets

Example 161. *Interpret the graph below*



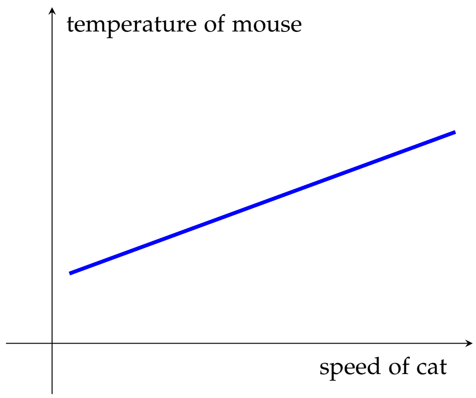
The more dolls there are, the _______________ each doll is

Example 162. *Interpret the graph below*



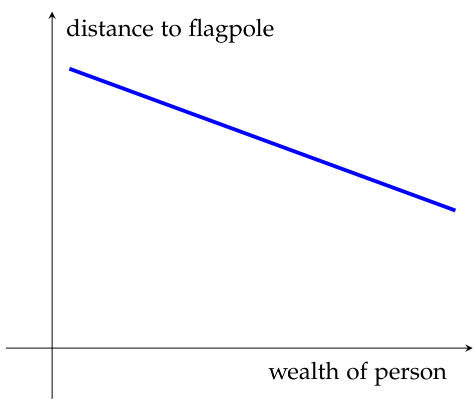
The higher a kite is, _______________

Example 163. *Interpret the graph below*



As the cat moves faster, _______________

Example 164. *Interpret the graph below*



Quadratic Equations

Our goal is to solve equations such as $5x^2 + 8x - 1 = 6$

Method 165: Zero Product Property

If $ab = 0$ then $a = 0$ or $b = 0$

Example 166. *Justify the Zero Product Property based on the geometric interpretation of multiplication.*

Example 167. *If $ab = 14$, can we conclude $a = 14$ or $b = 14$?*

Example 168. *Solve $x^2 - 8x + 15 = 0$*

Example 169. *Solve $x^2 - 3x = 0$*

Example 170. *Solve $x^2 - x = 6$*

Method 171: Zero Product Property, extended version

If $abc = 0$ then $a = 0$ or $b = 0$ or $c = 0$

Example 172. *Solve* $x^3 + 12x = 7x^2$

Method 173

1. Take a given equation and add/subtract on both sides to get one side to be 0.
2. If the other side factors, then apply the Zero Product Property. (If the other side does not factor, then try something else, since we're stuck.)

Example* 174. *Solve* $x^2 - 49 = 0$

Example 175. *Solve* $x^2 - 10x + 21 = 0$

Example 176. *Solve* $x^2 + 16 = 8x$

Example* 177. *Solve* $x^2 = 144$

Method 178: Square rooting both sides of an equation

Given an equation, make a copy of the equation one line below, and square root both sides, placing a $\pm$ sign on one side.

Note that simplifying a square root **expression** (See Concept85) does not involve $\pm$, but taking the action of square rooting both sides of an **equation** introduces $\pm$

Method 179: Quadratic equations without linear terms

To solve a quadratic equation with no linear term,

1. Add/subtract to get the quadratic term on one side and the constant term on the other side.
2. Square root both sides.

Though the first two questions questions on this page were on the previous page (Example 176 and Example 179), solve using the method introduced on this page.

Example 180. *Solve* $x^2 - 49 = 0$

Example 181. *Solve* $x^2 = 144$

Example 182. *Solve* $(x - 6)^2 = 19$

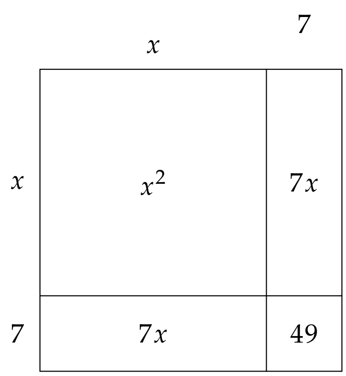
Example 183. *Solve* $(x+3)^2 = 5$

Example 184. *Geometrically justify* $(a+b)(c+d) = ac + ad + bc + bd$

Example 185. *Draw* $x \cdot x$, *also known as* x^2

Example 186. *Draw* $(x+y)(x+y)$, *and show the contribution of each term in the expanded form.*

Example 187. *Draw* $(x+5)(x+5)$, *and show the contribution of each term in the expanded form.*

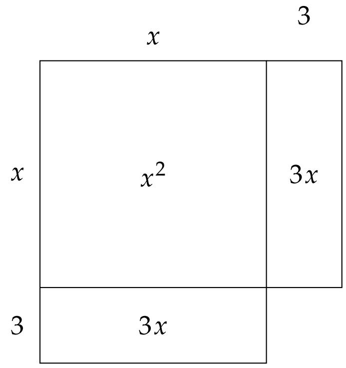


In the picture above, we see
$(x+7)^2 = (x+7)(x+7) = x^2 + 7x + 7x + 49 = x^2 + 14x + 49$

Concept 188

- The length of the second stick is 7
- The linear coefficient is 14, double the length of the second stick.
- The constant is 49, the square of the length of the second stick.

Example 189. *Draw* $x^2 + 20x + 100$, *but first rewrite the linear term.*



Example 190. *In the picture above, what is the missing amount that if added would complete the square?*

If you see $x^2 + 6x$, think of this as $x^2 + 3x + 3x$. Adding 9 to $x^2 + 3x + 3x$ would make a square.

Example 191. *What should be added to $x^2 + 16x$ to complete the square? Draw a picture of $x^2 + 16x$ and draw a picture of the completed square after adding.*

Example 192. *What should be added to $x^2 + 12x$ to complete the square? Draw the before and after pictures.*

Example 193. *Solve* $x^2 + 12x = 5$

Example 194. *Solve* $x^2 + 6x = 10$

Example 195. *Solve* $x^2 + 22x = -20$

$\boxed{\text{Linear coefficient}}$ divide by 2 $\rightarrow$ $\boxed{\text{Second stick}}$ square $\rightarrow$ $\boxed{\text{number to add}}$

Example 196. *Solve* $x^2 + 22x = -100$

Example 197. *Solve* $x^2 + 14x + 3 = 5$

Example 198. *Solve* $x^2 - 10x - 4 = 2$

Example 199. *Solve* $x^2 - 6x + 1 = 21$

Example 200. *Solve* $x^2 + 5x = 7$

Example 201. *Solve* $3x^2 + 5x = 6$

The last example showed us:

Method 202

Before computing second stick, etc. if the coefficient of x^2 is not 1, then divide both sides by the number that is in front of x^2 first.

Example 203. *Solve* $x^2 + 2x = 15$ *two different ways.*

Example 204. *Complete the square to solve for* x *in* $ax^2 + bx + c = 0$

.

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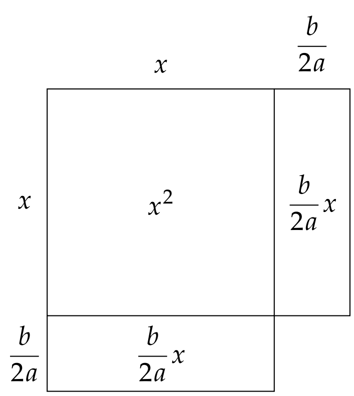
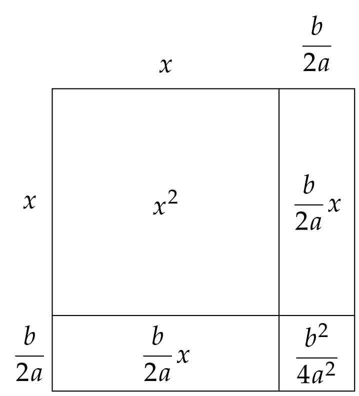
- The linear coefficient is $\frac{b}{a}$, not $\frac{b}{a}x$. Do not include the x.
- Since the linear coefficient is $\frac{b}{a}$, the length of the second stick is $\frac{b}{a} \div 2 = \frac{b}{a} \div \frac{2}{1} = \frac{b}{a} \cdot \frac{1}{2} = \frac{b}{2a}$
- Since the second stick is $\frac{b}{2a}$, the constant to add to both sides is $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{(2a)^2} = \frac{b^2}{2^2a^2} = \frac{b^2}{4a^2}$



Picture of $x^2 + \frac{b}{a}x$



Picture of $x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}$

Formula 205: Quadratic Formula

Given any quadratic equation $ax^2 + bx + c = 0$,
the two solutions of this equation are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We just obtained this formula (and you can see the work in the appendix on page 86).

The expression $\boxed{b^2 - 4ac}$ found inside the square root is called the *discriminant* and can be used to tell you the **number** of solutions.

- If $b^2 - 4ac < 0$, then the quadratic equation has 0 real solutions.
- If $b^2 - 4ac = 0$, then there is 1 real solution:
$$x = \frac{-b}{2a}$$
- If $b^2 - 4ac > 0$, there are 2 real solutions:
$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Example 206. *How many solutions does* $x^2 + 3x - 5 = 0$ *have?*

Example 207. *How many solutions does* $x^2 + x - 1 = 0$ *have?*

Example 208. *Solve* $x^2 + x - 1$. *Then, use the solutions to write* $x^2 + x - 1$ *in factored form.*

Example 209. *Solve* $\dfrac{2x}{x-2} + \dfrac{5}{x+2} = \dfrac{36}{x^2-4}$.

Completing the Square

Completing the square is a process that allows us to rewrite and solve quadratics. In this class, we will typically not use it to solve quadratics. Nevertheless, we will need it to rewrite quadratics to help us graph them and also find centers of circles. It is also the method used to derive the quadratic formula.

The idea behind completing the square are the following perfect squares:

- $x^2 + 2x + 1 = (x+1)^2$
- $x^2 + 4x + 4 = (x+2)^2$
- $x^2 + 6x + 9 = (x+3)^2$
- $x^2 + 8x + 16 = (x+4)^2$
- $x^2 + 10x + 25 = (x+5)^2$

Notice that I could try to use one of these to "factor" $x^2 + 10x + 26$. Now, I can rewrite 26 as 25+1 and then I have:

$$x^2 + 10x + 26 = x^2 + 10x + 25 + 1 = (x+5)^2 + 1.$$

Let's look at the following example. Suppose we wanted to solve $x^2 + 4x - 2 = 0$. Notice that I can add 0 without changing the expression:

$$x^2 + 4x + (0) - 2 = x^2 + 4x + (4 - 4) - 2$$

Now, group the terms that will give you a perfect square and factor:

$$(x^2 + 4x + 4) - 6 = (x + 2)^2 - 6.$$

This allows us to solve the quadratic equation:

$$x^2 + 4x - 2 = 0 \quad \Leftrightarrow \quad x^2 + 4x + 4 - 6 = 0 \quad \Leftrightarrow (x + 2)^2 - 6 = 0$$

We now solve using our regular techniques:

$$(x + 2)^2 = 6 \quad \Leftrightarrow \quad x + 2 = \pm\sqrt{6} \quad \Leftrightarrow \quad x = -2 \pm \sqrt{6}.$$

Example 210. *Solve* $x^2 + 3x - 5 = 0$ *using the quadratic formula.*

Example 211. *Solve* $x^2 + 3x - 5 = 0$ *by completing the square. Then, verify your last two answers are equivalent.*

Example 212. *Solve*[8] *for* s *in* $3.7 \times 10^{-6} = s(2s)^2$.

Example 213. *Solve*[9] *for* x *in* $8.1 \times 10^{-10} = \dfrac{x^2}{0.882}$

[8] This is the math behind a **molar solubility** question from **chemistry**.

[9] This equation is part of a chemistry problem.

Circles

Graphs of Circles

A **circle** with center (a, b) and radius r is the graph of an equation given by $(x - a)^2 + (y - b)^2 = r^2$. Notice that we can rewrite this as

$$r = \sqrt{(x - a)^2 + (y - b)^2}.$$

So, a circle consists of all points (x, y) that are a distance of r from the point (a, b).

Example 214. *Sketch the circle whose diameter has end points at* $(1, 5)$ *and* $(-3, 9)$.

Example 215. *Find the center and radius of the circle given by* $2x^2 + 2y^2 - 12x + 4y - 15 = 0$.

Example 216. *Find the center and radius of the circle given by* $x^2 + y^2 - 6x - 4y - 13 = 0$. *Then, sketch its graph.*

Example 217. *Graph the equation* $y = 2 + \sqrt{9 - (x - 1)^2}$.

Complex Numbers

Fact 218

$i = \sqrt{-1}$

Thus, $\sqrt{-4} = i \cdot \sqrt{4} = i \cdot 2 = 2i$. Taking the equation stated in the previous fact, we can square both sides to get

Fact 219

$i^2 = -1$

Example 220. *Simplify* $(3+4i)(5+2i)$

Example 221. *Solve* $x^2 + 25 = 0$

Example 222. *Solve* $x^2 + 6x = -52$

Solving additional equations

When you have an equation and there is a variable (like x or y), you can solve the equation for that variable by isolating the variable on one side of the equation and everything else to the other.

A solution to an equation is a value that when substituted in for the variable, makes the expression true. For example, 0 is not a solution to $x^2 - 1 = 0$ because $0^2 - 1 \neq 0$. And, 2 is not a solution to $1 = \frac{x-2}{x^2+x-6}$ because $1 \neq \frac{0}{0}$ and we cannot divide by 0). The expression $\frac{0}{0}$ is undefined. This is why it is super important to always check your work.

You can **always** check your work solving an equation, but here is one time you **must** check your work:

Habit 223

If the original equation had variables in the denominator, check to ensure there are no false solutions. (Any solution which causes division by zero to occur is a false solution.)

Example 224. *Find the solutions to* $\frac{6}{2x+11} + 5 = 5$

Example 225. *Find the solutions to* $\frac{4}{x+2} + \frac{1}{x-2} = \frac{5x-6}{x^2-4}$

Example 226. *Solve* $\frac{4}{2u-3} + \frac{10}{4u^2-9} = \frac{1}{2u+3}$.

Example 227. *Solve* $\frac{4}{5x+2} - \frac{12}{5x+6} = 0$.

Multiple variables

Example 228. *Solve for x in* $16 = 3x + 1$

Example 229. *Solve for x in* $a = 3x + 1$

Concept 230

We can still add/subtract/multiply/divide the same thing on both sides.

Example 231. *Solve for x in* $a = 3x + b$

Example 232. *Solve for x in* $a = cx + b$

Example 233. *Solve for s in* $as + bs + 8 = cs + d$

Recall Concept **??** about justifying collecting like terms.

Example 234. *Solve for w in* $A = 2\ell w + 2wh + 2\ell h$

Example 235. *Solve for α:* $\beta = \dfrac{\alpha}{1-\alpha}$

Example 236. *Solve* $\dfrac{1}{w} = \dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z}$ *for x.*

Example 237. *Solve*[10] *for r in* $F = G \cdot \frac{Mm}{r^2}$.

[10] In **physics**, the equation shown is Newton's Law of Gravitation.

Hidden Quadratics

A hidden quadratic is a type of equation that might not look like a quadratic, but can be solved using quadratic methods. For example: consider $x - \sqrt{x} - 6 = 0$. Notice that

$$x - \sqrt{x} - 6 = 0$$

is the same as

$$(\sqrt{x})^2 - \sqrt{x} - 6 = 0.$$

So, we have a *quadratic* in $\sqrt{x}$. To help you see this, let $y = \sqrt{x}$. Then,

$$(\sqrt{x})^2 - \sqrt{x} - 6 = 0$$

is the same as

$$y^2 - y - 6 = 0$$

which is an equation that we can solve:

$$(y-3)(y+2) = 0 \qquad \Rightarrow \qquad y = 3 \text{ and } y = -2$$

Remember that $y = \sqrt{x}$. So, we have $\sqrt{x} = 3$ and $\sqrt{x} = -2$. While $\sqrt{x} = 3$ makes sense, there is no solution to $\sqrt{x} = -2$.

Example 238. *Solve the following equation:* $2y^{1/3} - 3y^{1/6} + 1 = 0$

Example 239. *Solve the following equation:* $e^x + e^{-x} - 2 = 0$

Equations with multiple square roots

Consider the following example:

$$2\sqrt{x} - \sqrt{x-3} = \sqrt{5+x}$$

These are time intensive and require *a lot* of attention. And, as with hidden quadratics, we will always need to check our work. Here are some steps to follow:

- Isolate one of the square roots on one side of the equation

- Square both sides.
- Simplify and repeat until you don't have anymore square roots.
- Solve and check your answer in *original* equation

Habit 240

If the original equation had variables in a square root, check to ensure there are no false solutions. (Any solution which causes us to square root a negative is a false solution.)

Let's look at an example: Suppose we were asked to solve $2\sqrt{x} - \sqrt{x-3} = \sqrt{5+x}$

$$
\begin{aligned}
(2\sqrt{x} - \sqrt{x-3})^2 &= (\sqrt{5+x})^2 \\
(2\sqrt{x})^2 - 2(2\sqrt{x})(\sqrt{x-3}) + (\sqrt{x-3})^2 &= 5+x \\
4x - 4(\sqrt{x})(\sqrt{x-3}) + x - 3 &= 5+x \\
-4(\sqrt{x})(\sqrt{x-3}) &= 8-4x \\
(\sqrt{x})(\sqrt{x-3}) &= 2-x \\
(\sqrt{x}\sqrt{x-3})^2 &= (2-x)^2 \\
(\sqrt{x}\sqrt{x-3})^2 &= (2-x)^2 \\
(x)(x-3) &= 4-4x+x^2 \\
x^2 - 3x &= 4-4x+x^2 \\
x &= 4
\end{aligned}
$$

Remember to check your work:

$$2\sqrt{4} - \sqrt{4-3} = \sqrt{5+4} \Leftrightarrow 4-1=3$$

Equations using Factoring

Consider the equation: $y^{4/3} = 4y$. It may be tempting to divide by y. However, this requires that $y \neq 0$. Instead, a better way is to rewrite as $y^{4/3} - 4y = 0$, factor out a y:

$$y(y^{1/4} - 4) = 0,$$

and then solve each piece:

$$y = 0 \quad \text{and} \quad y^{1/4} - 4 = 0 \quad \Rightarrow \quad y = 0 \quad \text{and} \quad y = 64$$

Example 241. *Solve the following equation:* $\sqrt{3-x} - x = 3$

Example 242. *Solve the following equation:* $6x^{-1/2} - 13x^{-1/4} + 6 = 0$

Example 243. *Solve the following equation:* $2\sqrt{x} - \sqrt{x-3} = \sqrt{5+x}$

Example 244. *Solve the following equation:* $\sqrt{x+3} = \sqrt[4]{2x+6}$

Inequalities

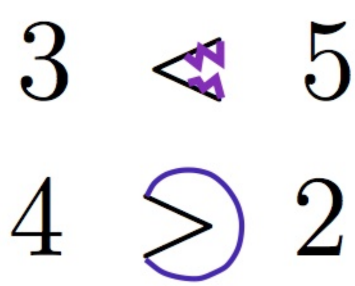
Inequalities

Inequalities are a way to order numbers. In other words, they are a way to know when one number is bigger, smaller, or equal to another number. We use the following symbols:

- $<$ is read "less than"
- $\leq$ is read "less than or equal to"
- $>$ is read "greater than"
- $\geq$ is read "greater than or equal to"

For example: $2 < 9$ and $-5 \geq -6$. Notice, though, that $x \geq y$ is equivalent to $y \leq x$

One way to remember which way the inequality goes is to "eat the biggest Number"



Here is some terminology:

- $a > b$ means a is greater than b.
- $a \geq b$ means a is greater than or equal to b.
- $a < b$ means a is less than b.
- $a \leq b$ means a is less than or equal to b.

Some things to remember with inequalities are that if $0 < a < b$, then $\frac{1}{a} > \frac{1}{b}$. Also, when you multiply or divide by negative numbers, the inequality sign switches. Can you think of why this is the case?

Interval Notation

We can use inequalities to describe sets of numbers. For example, if we were to write $1 < x < 3$, the x represents any number that is greater than 1 and less than 3. While there is nothing wrong with using inequalities to describe sets of numbers, we can also use a shorthand method called interval notation. Interval notation is preferred in college-level mathematics. Below are the corresponding inequalities and intervals:

- $a < x < b$ corresponds to (a, b)

- $a \leq x < b$ corresponds to $[a, b)$

- $a < x \leq b$ corresponds to $(a, b]$

- $a \leq x \leq b$ corresponds to $[a, b]$

- $a \leq x$ corresponds to $[a, \infty)$

- $x \leq b$ corresponds to $(-\infty, b]$

Here are some comments:

- The smallest number always goes to the left and the largest number always goes to the right. For example, $(2, 8)$ is okay , but $(8, 2)$ is not okay.
- If you **DON'T** include the endpoint, use either "(" or ")"

- If you **DO** include the endpoint, use either "[" or "]"
- ∞ and $-\infty$ always use "(" and ")"
- This is what WeBWork wants you to use (unless stated otherwise)
- If you have multiple intervals, you use a "U" to combine them:
  $-1 \leq x < 4$ or $x > 9$ is represented by

$$[-1, 4) \cup (9, \infty)$$

Example 245. *What is the interval notation that corresponds to the following set of numbers?*

- $x > 1$ *or* $x < -1$

- $x \leq 1$ *or* $x > -1$

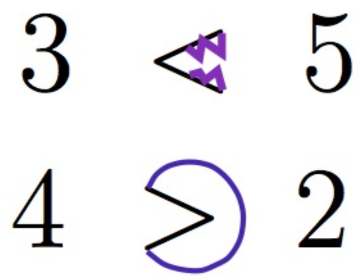
- *All real numbers, x, so that* $x \neq 1, 2$

- *x so that* $x^2 \neq 2$

Review

Recall the following facts about inequalities:

- $x \geq y$ is equivalent to $y \leq x$
- If $0 < a < b$, then $\frac{1}{a} > \frac{1}{b}$
- If $a < b$, then $-a > -b$
- Eat the biggest number



Example 246. *Find all x so that* $2x + 5 < 4$. *Provide your answer as both an inequality and in interval notation.*

Example* 247. *Find all x so that* $2x + 10 \geq 0$. *Provide your answer as both an inequality and in interval notation.*

Example 248. *Find all x so that* $-3 < -2x + 5 \leq 5$. *Provide your answer as both an inequality and in interval notation.*

Example 249. *Find all x so that* $4x - 3 \leq 2x + 5$. *Provide your answer as both an inequality and in interval notation.*

Example 250. *Find all x so that* $-6 < 2x - 4 \leq 2$. *Provide your answer as both an inequality and in interval notation.*

Example 251. *Find all x so that* $-5 \leq \frac{4 - 3x}{2} < 1$. *Provide your answer as both an inequality and in interval notation.*

Example 252. *Find all x so that* $9 + \frac{1}{3}x \geq 4 - \frac{1}{2}x$. *Provide your answer as both an inequality and in interval notation.*

Example 253. *Find all x so that* $4 > \frac{2 - 3x}{7} \geq -2$. *Provide your answer as both an inequality and in interval notation.*

Absolute Value Equations

Example 254. *Solve* $|x| = 5$, *with final answer not mentioning absolute value.*

Example 255. *Solve* $|u| = 6$*, with final answer not mentioning absolute value.*

Example 256. *Solve* $|x| - 2 = 5$

Example 257. *Solve* $|x - 2| = 5$

Warning 258

Taking $|x - 2| = 5$ and adding 2 on both sides does not lead to $|x| = 7$. This instead leads to $|x - 2| + 2 = 7$, and unfortunately, the -2 inside the absolute value and the $+2$ outside the absolute value don't interact.

Answer: Replace every $\boxed{x-2}$ with $\boxed{u}$

$$|u| = 5$$

State the values of u which satisfy this equation.

$$u = -5 \quad \text{or} \quad u = 5$$

Replace every $\boxed{u}$ with $\boxed{x-2}$

$$x - 2 = -5 \quad \text{or} \quad x - 2 = 5$$

Solve each equation independently

$$x = -3 \quad \text{or} \quad x = 7$$

Example 259. *Solve* $|x + 8| = 10$

Example 260. *Solve* $|3x - 4| = 5$

Example 261. *Solve* $3|4x + 1| = -3$

Example 262. *Solve* $\left|\frac{x}{2} - \frac{2}{3}\right| = 4$

Example 263. *Solve* $\left|\frac{x}{3} - 12\right| = 10$

Example 264. *Solve for* x *in*

$$\left(\frac{|3(x^4 - x^2) + 1| - 4}{5} - 9\right)^2 + 4\left(\frac{|3(x^4 - x^2) + 1| - 4}{5} - 9\right) = 21$$

Answer: Let $u = \dfrac{|3(x^4 - x^2) + 1| - 4}{5} - 9$.

$$u^2 + 4u = 21$$

$$u^2 + 4u - 21 = 0$$
$$(u+7)(u-3) = 0$$
$$u + 7 = 0 \text{ or } u - 3 = 0$$
$$u = -7 \text{ or } u = 3$$
$$\frac{|3(x^4 - x^2) + 1| - 4}{5} - 9 = -7 \quad \text{or} \quad \frac{|3(x^4 - x^2) + 1| - 4}{5} - 9 = 3$$

$$\frac{|3(x^4 - x^2) + 1| - 4}{5} = 2 \quad \text{or} \quad \frac{|3(x^4 - x^2) + 1| - 4}{5} = 12$$
$$|3(x^4 - x^2) + 1| - 4 = 10 \quad \text{or} \quad |3(x^4 - x^2) + 1| - 4 = 60$$
$$|3(x^4 - x^2) + 1| = 14 \quad \text{or} \quad |3(x^4 - x^2) + 1| = 64$$

Let $m = 3(x^4 - x^2) + 1$

$$|m| = 14 \quad \text{or} \quad |m| = 64$$

$$m = 14 \text{ or } m = -14 \quad \text{or} \quad m = 64 \text{ or } m = -64$$

$$3(x^4 - x^2) + 1 = 14 \text{ or } 3(x^4 - x^2) + 1 = -14 \text{ or } 3(x^4 - x^2) + 1 = 64 \text{ or } 3(x^4 - x^2) + 1 = -64$$

$$3(x^4 - x^2) = 13 \text{ or } 3(x^4 - x^2) = -15 \text{ or } 3(x^4 - x^2) = 63 \text{ or } 3(x^4 - x^2) = -65$$

$$x^4 - x^2 = \frac{13}{3} \text{ or } x^4 - x^2 = -5 \text{ or } x^4 - x^2 = 21 \text{ or } x^4 - x^2 = -\frac{65}{3}$$

$$(x^2)^2 - x^2 = \frac{13}{3} \text{ or } (x^2)^2 - x^2 = -5 \text{ or } (x^2)^2 - x^2 = 21 \text{ or } (x^2)^2 - x^2 = -\frac{65}{3}$$

Let $s = x^2$

$$s^2 - s = \frac{13}{3} \text{ or } s^2 - s = -5 \text{ or } s^2 - s = 21 \text{ or } s^2 - s = -\frac{65}{3}$$

Now, solve these four quadratic equations (each by completing the square). Once values of s are found, go replace s with x^2.

Absolute Value Inequalities

Example 265. *Solve* $|x| \leq 3$*, stating an answer without absolute values.*

Example 266. *Solve* $|x| \geq 3$*, stating an answer without absolute values.*

Example 267. *Solve* $|x| < 3$*, stating an answer without absolute values.*

Example 268. *Solve* $|r| > 3$*, stating an answer without absolute values.*

Example* 269. *Solve* $|u| \leq 7$*, stating an answer without absolute values.*

Example 270. *Solve* $|x| + 3 \leq 10$

Example 271. *Solve* $|x + 3| \leq 10$

Warning 272

Taking $|x+3| \leq 10$ and subtracting 3 on both sides does not lead to $|x| \leq 7$. This instead leads to $|x+3| - 3 \leq 7$, and unfortunately, the $+3$ inside the absolute value and the -3 outside the absolute value don't interact.

How can we use Example 271?

Example 273. *Solve* $|3x-4| < 5$

Example* 274. *Solve* $|x-2| \geq 10$

Example 275. *Solve* $|x-2| + 2 \geq 10$

Example 276. *Solve* $-4|4x+1| \leq -3$

Example 277. *Solve* $\left|\frac{x}{2} - \frac{2}{3}\right| \geq -4$

Example 278. *Solve* $\left|12 - \frac{x}{3}\right| < 10$

Functions

Functions

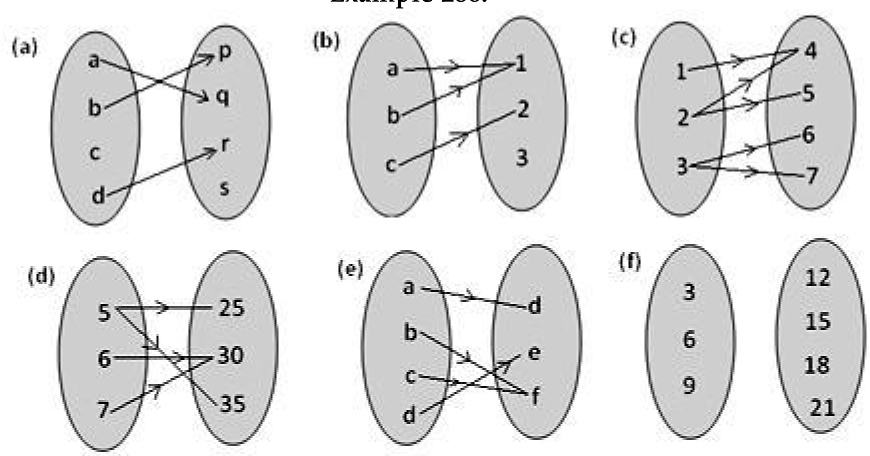
Definition 279: Function

A function is a rule that describe exactly one output for each input.

Each input has one output. We cannot have an input with two outputs, and we cannot have an input with no outputs.

NO YES NO

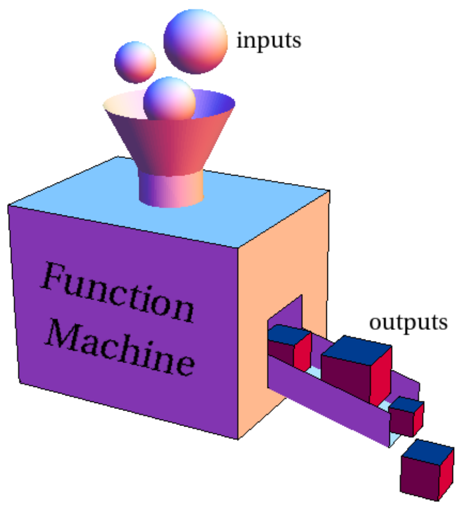
Example 280.



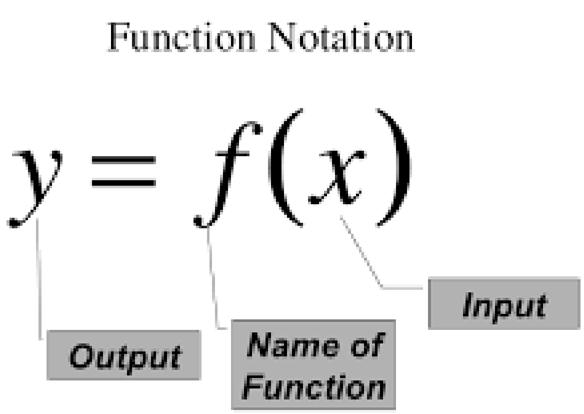
NO YES NO

A function consists of three things:

- A set of inputs (stuff that goes into the function) called the **domain**
- A set of outputs (stuff that comes out of the function) called the bf range
- A rule assigning each input to exactly one output



We have specific notation for functions:



Example 281. *If* $f(x) = x^2 + 6$, *find* $f(10)$

Answer: $f(10) = (10)^2 + 6$.

After simplifying, we have $f(10) = 100 + 6$ and simplifyng further, $f(10) = 106$.

Language Discussion 282

Use the words input and output when talking about a function.

Example 283. *For the function* $f(x) = x^2 + 6$,
if the input is 10*, the output is* 106.
This is summarized by writing $f(10) = 106$.

Example 284. *For the function* $f(x) = x^2 + 6$,
if the input is 5*, the output is* 31.
This is summarized by writing $f(5) = 31$.

Language Discussion 285: Function

When seeing/writing $f(x)$ say "f of x" and do not skip the word of. Similarly, $f(5)$ should be said "f of 5". If you skip the of, it leads to the error of thinking we have "f times 5" but there is no multiplication going on between f and 5 at all!

Example 286. *Given* $f(x) = 10^x + 6x$ *find* $f(2)$

Answer: $f(2) = 10^2 + 6(2)$. Simplified, $f(2) = 112$. In words, the input 2 leads to the output 112.

Warning 287

In the last example, we wrote $\boxed{f(2) = 10^2 + 6(2)}$ which is correct.
Writing $\boxed{f(x) = 10^2 + 6(2)}$ is incorrect.

Method 288: Finding the output for an input

Task: $f(x) = 10^x + 6x$ find $f(2)$.

How to restate the task: Notice the 2 inside parentheses.
While writing $\boxed{f(x) = 10^x + 6x}$ replace every $\boxed{x}$ you see with $\boxed{2}$. This even applies to the x that is on the left of the equal sign.

Replace every $\boxed{\text{vowel}}$ in $\boxed{\text{your name}}$ with $\boxed{\text{oodle}}$

Given $f(x) = 10^x + 6x$ to find $f(2)$ you must:
Replace every $\boxed{}$ in $\boxed{}$ with $\boxed{}$

Given $f(x) = 10^x + 6x$ to find $f(c)$ you must:
Replace every $\boxed{}$ in $\boxed{}$ with $\boxed{}$

Given $f(x) = x^2 + x + 3$ to find $f(a + h)$ you must:
Replace every $\boxed{}$ in $\boxed{}$ with $\boxed{}$

Given $f(x) = x^2 + x + 3$ to find $f(x + h)$ you must:

Replace every $\boxed{}$ in $\boxed{}$ with $\boxed{}$

Example 289. *If* $f(x) = x^2 + x + 3$*, find* $f(3)$*,* $f(c)$*, and* $f(x+h)$

Notice that if you are asked to find $f(x+h)$, *all* of $x+h$ is put into the function.

Example 290. *If* $f(x) = x^2$*, find* $f(x+h)$ *and* $f(x) + h$*.*

Warning 291: Function does not "distribute"

$f(x+h) \neq f(x) + f(h)$

Example 292. *Given the function* $f(x) = 3x + 1$*, find* $\dfrac{f(x+h) - f(x)}{h}$*.*

We see that

$$f(x+h) = 3(x+h) + 1 \quad \text{and} \quad f(x) = 3x + 1.$$

Therefore,

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{3(x+h) + 1 - (3x+1)}{h} \\ &= \frac{3x + 3h + 1 - 3x - 1}{h} \\ &= \frac{3h}{h} \\ &= 3 \end{aligned}$$

Example 293. *Given the function* $f(x) = \dfrac{1}{x}$*, find* $\dfrac{f(x+h) - f(x)}{h}$

$$f(x+h) = \frac{1}{x+h} \quad \text{and} \quad f(x) = \frac{1}{x}$$

So,

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ &= \frac{1}{h}\left(\frac{x}{x(x+h)} - \frac{x+h}{x(x+h)}\right) \\ &= \frac{-1}{x(x+h)} \end{aligned}$$

Example 294. *Given the function* $f(x) = -10x^2 + 3x$*, find* $\dfrac{f(x+h) - f(x)}{h}$ *and simplify*[11]*.*

[11] In **business**, $f(x) = -10x^2 + 3x$ is typical of a **revenue function** looks. The expression we found is called the **average revenue**

Domains

The set of numbers that make sense in the function is called the domain. Unless the domain is given to you, assume it's as large as possible. For example, if $f(x) = \sqrt{x}$, $f(-2) = \sqrt{-2}$ doesn't make sense so -2 is not in the domain of this function.

Method 295: How to find domain

- Avoid diving by zero. (In other words, what values of x cause the denominator to be zero? We must exclude these x-values.)
- Avoid square rooting a negative number. Rephrased, we want/need what we are square rooting to be greater than or equal to 0. In other words, we need the content *inside* a square root to be ≥ 0.

Example 296. *Find the domain of* $f(x) = \sqrt{2x+10}$

Answer: We need $2x + 10 \geq 0$. In Example 249, we saw the answer to this was $[-5, \infty)$. So the domain of f is $[-5, \infty)$.

Example 297 (Follow up)**.** *If* $f(x) = \sqrt{2x+10}$ *find* $f(-3)$

Note: Though the x-value $x = -3$ is negative, we are not square rooting x itself. We are square rooting $2x + 10$. We already note that the domain was found to be $[-5, \infty)$, and -3 is in this interval.

Example 298. *Find the domain of* $f(x) = \sqrt{|x-2| - 10}$

Answer: We need $|x - 2| - 10 \geq 0$. So $|x - 2| \geq 10$. We solved this inequality in Example 276 and got $(-\infty, -8) \cup (12, \infty)$ as our answer. Thus, the domain of f is $(-\infty, -8) \cup (12, \infty)$.

Example 299. *Find the domain of* $f(x) = \dfrac{x^2+5x+6}{x^2+2x-24}$

Example 300. *Find the domain of* $f(x) = \dfrac{2x+\sqrt{3x+21}}{x^2+2x-24}$

Graphs of Functions

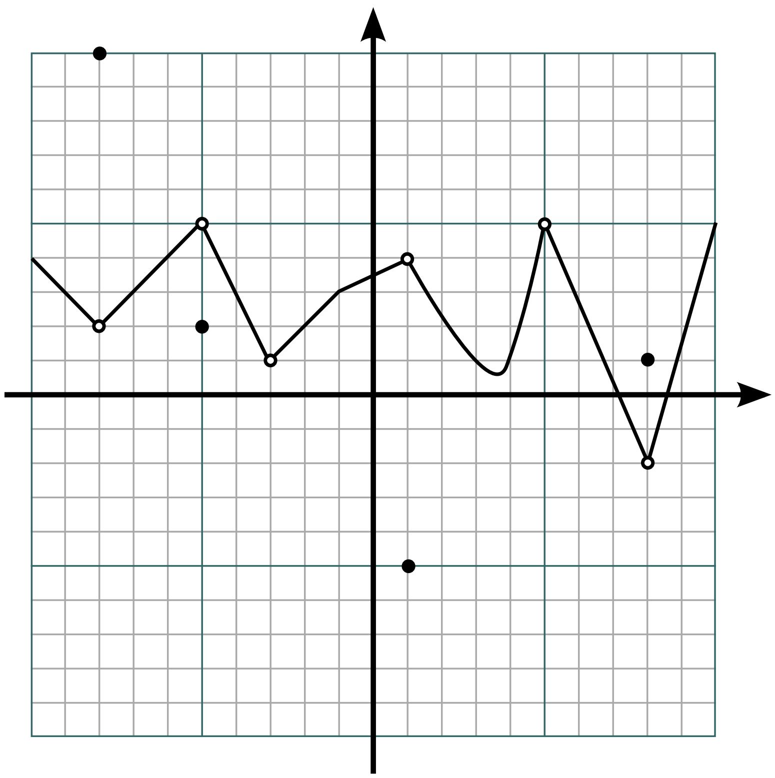
Given a function f, a point on the graph of f is given by $(x, f(x))$. The x-axis is the "input" axis and the y-axis is the "output" axis. In other words:

Concept 301

$f(a) = b$ if and only if the point (a, b) is on the graph of f

Lots of graphs do not represent functions. A graph must pass the Vertical Line Test (VLT) in order to be a graph of a function. The Vertical Line Test states that every vertical line must pass through the graph at *most* one time.

Example 302. *Find three points on the graph of* $f(x) = x^2 + 6$



Example 303. *The graph above passes the Vertical Line Test, so let us say that this graph introduces to us a function* $y = f(x)$*. Find* $f(6)$*.*

Note: we have done this task using different language in Example 155.

Linear Functions

A **linear function** is a function of the form:

$$f(x) = mx + b.$$

The value m is the slope of the line and b is the y-intercept.Here, the $f(x)$ is the output just like y in the equation $y = mx + b$.

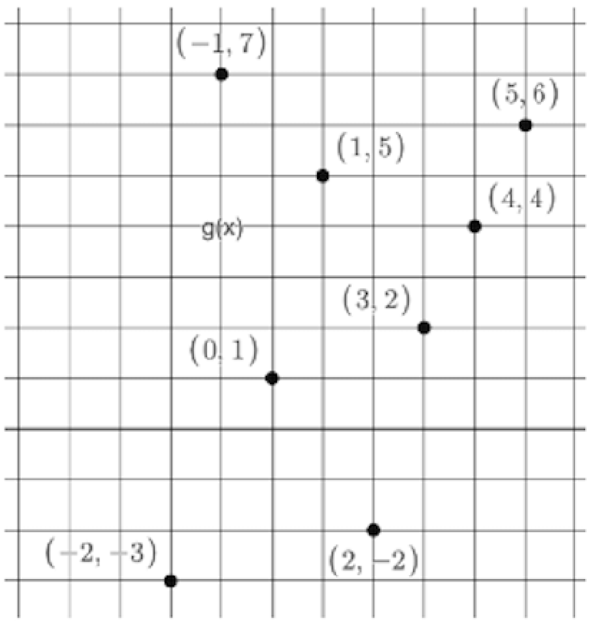
Example 304. *Let* f *be a linear function given by* $f(x) = 3x + 5$*. Find* $f(0), f(2), f(-10)$

Example 305. *Determine if the function* $f(x) = \dfrac{x+1}{5}$ *is a linear function. If so, identify the slope and y-intercept.*

Example 306. *Determine the point of intersection of the graphs of the functions* $f(x) = 3x + 1$ *and* $g(x) = -2x + 5$.

Example 307. *A pond is being filled at a rate of 10 gallons per minute. Initially the pond contains 300 gallons. Find a linear function that models this situation. Then, determine when the pond will contain 1300 gallons.*

Example 308. *Find a linear function whose graph passes through* $(3, -2)$ *and* $(2, 0)$.



x	$h(x)$
-3	5
-2	3
-1	1
0	-1
1	-3
2	-5
3	-7

The picture in the upper left shows the graph of a function called $g(x)$ while the table on the right shows the inputs/outputs for a function called $h(x)$.

Example 309. *Find* $g(2)$

Example 310. *Find* $h(3)$

Example 311. *Find* $h(3) + g(5)$

Example 312. *Find* $h(1) + g(2) + 10^{g(3)} + 10\,h(-3)$

Example 313. *For what value(s) of x is* $g(x)$ *equal to* 2*? How is this question different from the first question on this page?*

Quadratic Functions

A **Quadratic Function** is a function whose rule is a quadratic expression. One such example would be:

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

There are many ways to represent a quadratic expression. Here is another example of a quadratic function:

$$f(x) = a(x-h)^2 + k.$$

This last example is called the vertex form of a quadratic function. Every quadratic function can be written in both standard and vertex form.

Example 314. *Rewrite the function $f(x) = (x-2)^2 + 1$ in standard form*

Example 315. *Rewrite the function $f(x) = -3(x+1)^2 - 7$ in standard form.*

Standard Form to Vertex Form

Rewriting a quadratic function from standard form to vertex form is a bit more challenging. However, the good news is that there is a formula we can use. This will be given at the end of this section. Suppose we are given the quadratic function $f(x) = ax^2 + bx + c$. We want to change this to $y = ax^2 + bx + c$,subtract c,and divide by a.This gives you:

$$\frac{y-c}{a} = x^2 + \frac{b}{a}x.$$

We do this so that we can complete the square. We do this by adding $\left(\frac{b}{2a}\right)^2$ to both sides:

$$\frac{y-c}{a} + \left(\frac{b}{2a}\right)^2 = x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2$$

$$\frac{y-c}{a} + \left(\frac{b}{2a}\right)^2 = \left(x + \frac{b}{2a}\right)^2$$

Now, we want to solve for y:

$$\frac{y-c}{a} = \left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2$$

$$y - c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a}$$

$$y = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

Our last step is to write y as $f(x)$:

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

You will; notice that given a standard form of a quadratic function, $f(x) = ax^2 + bx + c$, the corresponding vertex form is:

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

Example 316. *Write $f(x) = -x^2 - 4x + 5$ in vertex form.*

Example 317. *Rewrite the function $f(x) = x^2 + 8x - 2$ in vertex form.*

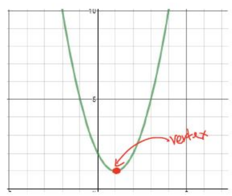
Example 318. *Rewrite the function $f(x) = 2x^2 + 3x - 1$ in vertex form.*

Graphing Quadratic Functions

$$f(x) = ax^2 + bx + c \qquad\qquad f(x) = a(x-h)^2 + k$$

Vertex Form

When working with quadratic functions, we prefer to use the vertex form for many reasons such as it allows us to find the range and min/max values. Given a quadratic in vertex form $f(x) = a(x - h)^2 + k$, the vertex is the point (h, k).



We also know that

- if $a < 0$, then the quadratic opens downwards,
- if $a > 0$, then the quadratic opens upwards.

Example 319. *Sketch the graph of $f(x) = (x-2)^2 + 1$. Then, find the domain and range of $f(x)$.*

Example 320. *Sketch the graph of $y = -x^2 - 3$. Then, find the domain and range of y.*

Example 321. *Sketch the graph of $f(x) = x^2 + 2x + 5$. Then, find the domain and range of $f(x)$.*

Standard Form to Vertex Form

Given a quadratic in standard form, $f(x) = ax^2 + bx + c$, the vertex is given by

$$(h,k) = \left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$$

This allows you to write a standard quadratic in vertex form without completing the square. The vertex is a point on the graph of a function, if you can remember the x-coordinate is $\frac{-b}{2a}$, then you can find the y-coordinate by putting this x-value into the function f to get a y-value.

Example 322. *Write $f(x) = -x^2 + 4x + 5$ in vertex form. Graph the function $f(x)$. Then state which x-value leads to the largest y-value*[12].

[12] In business, the function $f(x) = -x^2 + 4x + 5$ is the typical look of a **profit function**. Finding the point on the graph which has the largest y-value is maximizing the profit of a business.

Example 323. *Graph the function* $f(x) = x^2 - 4x + 5$*. Then, find the minimum of* $f(x)$*.*

Example 324. *Find the x and y intercepts of* $f(x) = x^2 - 4x + 5$*.*

Example 325. *Graph the function* $h(x) = -4x^2 + 9x + a$*. Determine the maximum of* $g(x)$*.*

Example 326. *Find the quadratic that has vertex* $(2, 4)$ *and passes through the point* $(1, 2)$*.*

Example 327. *Find the quadratic with a minimum* $y = 3$*, that is symmetric about the line* $x = 1$ *and passes through the point* $(5, 9)$*.*

Let's suppose that we want to find x so that $x^2 - 4x - 15 \leq 6$. We can use technology to see the values of x that satisfy this inequality. However, by rewriting $x^2 - 4x - 15 < 6$ as $x^2 - 4x - 21 < 0$, we can factor this quadratic to get

$$(x - 7)(x + 3) < 0$$

It is tempting to solve this similar to a quadratic, however we must use a sign chart and test values.

Method 328: Solving a polynomial inequality

- Set one side of the inequality to 0
- Find the values of x that make the quadratic equal to 0
- Put these numbers on a number line
- Find numbers in each of the regions
- Plug the test values into the quadratic
- Determine when the quadratic is positive or negative
- Choose the region that satisfies the inequality.
- If there are multiple separate intervals, write down intervals using $\cup$

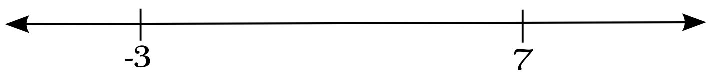
Example 329. • *Set one side of the inequality to* 0

$$x^2 - 4x - 21 < 0$$

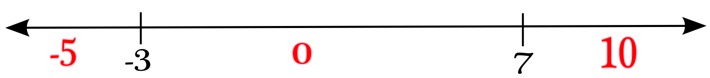
- *Find the values of x that make the quadratic equal to* 0

$$(x - 7)(x + 3) = 0 \leftrightarrow x = 7, \quad x = -3$$

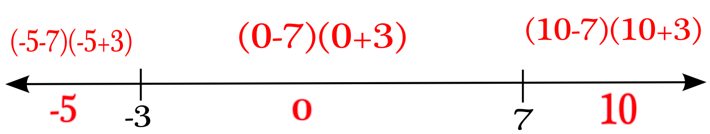
- *Put these numbers on a number line*



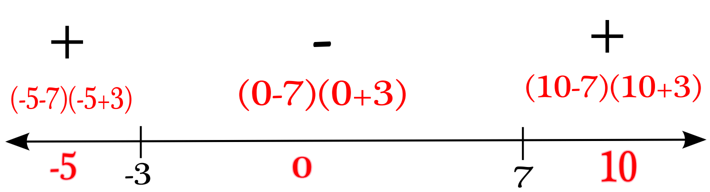
- *Find numbers in each of the regions*



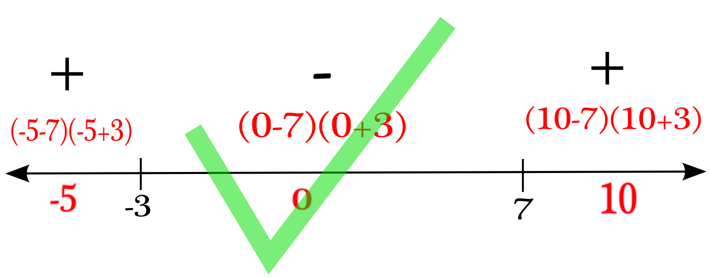
- *Plug the test values into the quadratic*



- *Determine when the quadratic is positive or negative*



- *Choose the region that satisfies the inequality*



- *For this problem, the answer is* $(-3, 7)$.

Example 330. *Solve* $x^2 + 4x + 3 \leq 0$

Example 331. *Solve* $x(3x - 1) \geq 4$

Example 332. *Solve* $2x^3 - 3x^2 - 2x + 3 \leq 0$.

Polynomial functions, zeroes

A polynomial function is a function of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \ldots a_2 x^2 + a_1 x^1 + a_0,$$

where the a_i's are constants and the exponents, n, are non-negative integers. The largest exponent of x is called the **degree** of the polynomial f. The coefficient (ie the number) in front of the term with the highest exponent for x is called the **leading coefficient** for the polynomial f.

Example 333. *Identify the degree and leading coefficient for the polynomials* $f(x) = 27x^4 + \sqrt{\pi}x - 100x^{10}$, $g(x) = x + 2$ *and* $h(x) = x^5 + x^4 + x^3 + x^2 + x^1 + 1$.

Suppose that the polynomial is in factored form:

$$f(x) = a(x - r_1)^{k_1}(x - r_2)^{k_2} \ldots (x - r_j)^{k_j}$$

We can easily graph these polynomials know the r_i's, leading coefficient, and degree.

Factors of Polynomials

Suppose that the polynomial is in factored form:

$$f(x) = a(x - r_1)^{k_1}(x - r_2)^{k_2} \ldots (x - r_j)^{k_j}$$

We know that r_i are roots. This means:

- $(r_i, 0)$ is on the graph of f (r_i are x-intercepts)
- $f(r_i) = 0$

The corresponding exponent, k_i, for the factor $(x - r_i)$ is called the **multiplicity** of the root r_i.

Example 334. *Is $x + 4$ a factor of $x^3 + 64$? Find all roots with multiplicities of $x^3 + 64$.*

Example 335. *Give an example of a polynomial of degree 7 with roots at $x = -3, 0, 1, 5$.*

Terminology

Let's go over some vocabulary. The words **zero**, **root**, and **x-intercept** all mean the same thing: Given a polynomial $p(x)$, a zero is *any* value of x so that $p(x) = 0$. A zero is also *any* value of x where the graph of $p(x)$ touches/crosses the x-axis.

Graphing Polynomials

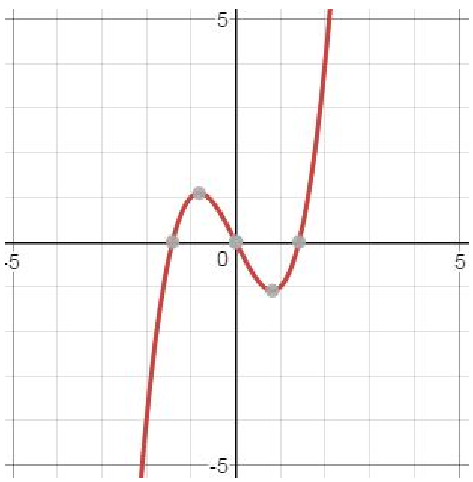
Vocabulary of Polynomials

A polynomial function is a function of the form:

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots a_2 x^2 + a_1 x^1 + a_0,$$

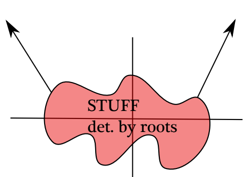
where the a_i's are constants and the exponents, n, are non-negative integers. The largest exponent of x is called the **degree** of the polynomial f. The coefficient (ie the number) in front of the term with the highest exponent for x is called the **leading coefficient** for the polynomial f.

Polynomials are **continuous functions**. This means that you can draw it without picking up your pencil.

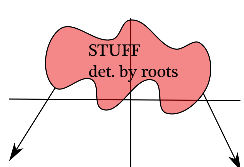


Degree of Polynomials

Suppose that the degree of a polynomial is *even*, then we can say something about the ends of the graph:

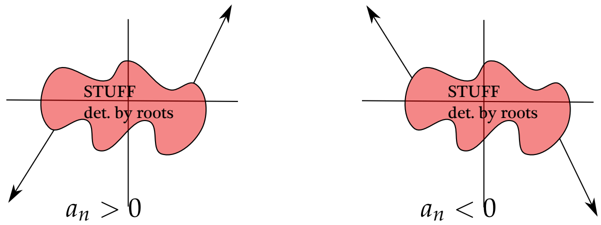


$a_n > 0$



$a_n < 0$

Suppose that the degree of a polynomial is *odd* then we can say something about the ends of the graph: Here, the a_n represents the leading coefficient for f.



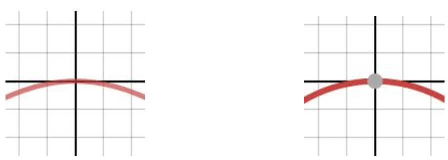
Factors of Polynomials

Suppose that the polynomial is in factored form:

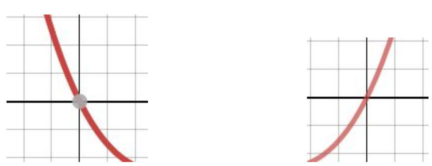
$$f(x) = a(x - r_1)^{k_1}(x - r_2)^{k_2} \dots (x - r_j)^{k_j}.$$

We know that r_i are roots. This means the points $(r_i, 0)$ are on the graph of f (r_i are x-intercepts) and $f(r_i) = 0$. The corresponding exponent, k_i, for the factor $(x - r_i)$ is called the **multiplicity** of the root r_i.

It turns out that if r_i has multiplicity that is *even*, then the graph looks like teh graph below near the root r_i. In other words, the graph touches, but does not cross the x axis.



However, if r_i has multiplicity that is *odd*, then the graph crosses the x axis near r_i, similar to the graph below.



Degrees of Polynomials

Suppose that the polynomial is in factored form:

$$f(x) = a(x - r_1)^{k_1}(x - r_2)^{k_2} \dots (x - r_j)^{k_j}.$$

We can find the degree of the polynomial f by summing up all of the multiplicities:

$$k_1 + k_2 + k_3 + \dots k_j.$$

The leading coefficient for the polynomial f is the number a.

Example 336. *Sketch the graph of $f(x) = x^2(x - 1)(x - 3)^2$*

Example 337. *Sketch the graph of* $f(x) = -x^2 - 4x^4$

Example 338. *Sketch the graph of* $f(x) = (x^2 - 2)(x + 7)(x - 4)$

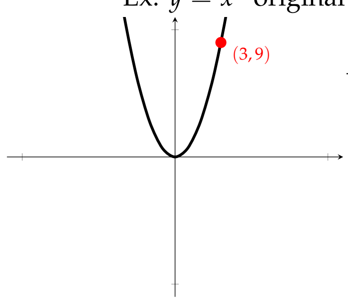
Graph Transformations

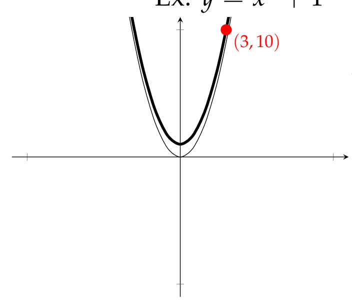
Graph Transformations

Given a function f, a point on the graph of f is given by $(x, f(x))$. The x-axis is the "input" axis and the y-axis is the "output" axis.

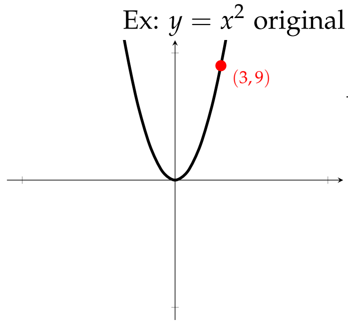
The old method of graphing: take different values of x and find the corresponding y ($f(x)$) values, plot points, then sketch a graph. This is the least preferred method because it takes too much time and is not accurate. The new method is to use base graphs and modify them accordingly

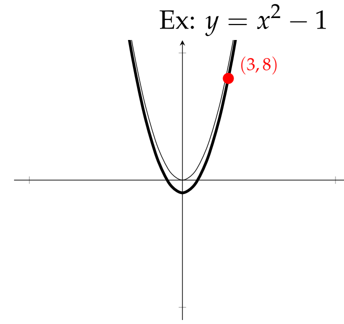
Vertical transformations
Our example uses $f(x) = x^2$. Because (a, b) is on the graph of $y = f(x)$ if and only if $f(a) = b$, we get...

<table>
<tr>
<td>$y = f(x)$
vs.
$y = f(x) + c$</td>
<td>$y = f(x)$ original
Ex: $y = x^2$ original


<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr><tr><td>4</td><td>16</td></tr><tr><td>5</td><td>25</td></tr><tr><td>6</td><td>36</td></tr></table></td>
<td>shift up by c
Example:
shift up by 1</td>
<td>$y = f(x) + c$
Ex: $y = x^2 + 1$


<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>5</td></tr><tr><td>3</td><td>10</td></tr><tr><td>4</td><td>17</td></tr><tr><td>5</td><td>26</td></tr><tr><td>6</td><td>37</td></tr></table></td>
</tr>
</table>

<table>
<tr>
<td>$y = f(x)$
vs.
$y = f(x) - c$</td>
<td>$y = f(x)$ original
Ex: $y = x^2$ original


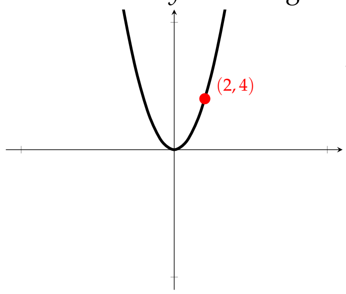
<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>9</td></tr><tr><td>4</td><td>16</td></tr><tr><td>5</td><td>25</td></tr><tr><td>6</td><td>36</td></tr></table></td>
<td>shift down by c
Example:
shift down by 1</td>
<td>$y = f(x) - c$
Ex: $y = x^2 - 1$


<table><tr><th>x</th><th>y</th></tr><tr><td>0</td><td>-1</td></tr><tr><td>1</td><td>0</td></tr><tr><td>2</td><td>3</td></tr><tr><td>3</td><td>8</td></tr><tr><td>4</td><td>15</td></tr><tr><td>5</td><td>24</td></tr><tr><td>6</td><td>35</td></tr></table></td>
</tr>
</table>

$y = f(x)$ vs. $y = c\,f(x)$

$y = f(x)$ original

Ex: $y = x^2$ original

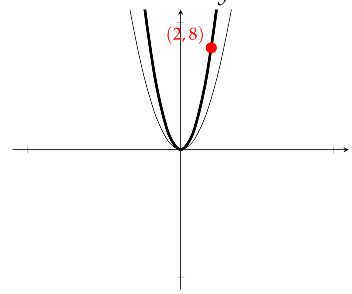


x	y
0	0
1	1
2	4
3	9
4	16
5	25
6	36

vert stretch by c
Example:
vert stretch by 2

$y = c\,f(x)$

Ex: $y = 2x^2$

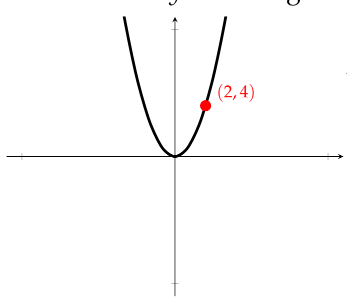


x	y
0	0
1	2
2	8
3	18
4	32
5	50
6	72

$y = f(x)$ vs. $y = \frac{1}{c} f(x)$

$y = f(x)$ original

Ex: $y = x^2$ original

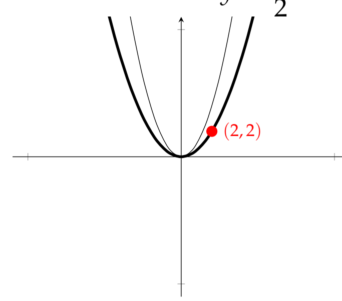


x	y
0	0
1	1
2	4
3	9
4	16
5	25
6	36

vert shrink by c
Example:
vert shrink by 2

$y = \frac{1}{c} f(x)$

Ex: $y = \frac{1}{2}x^2$

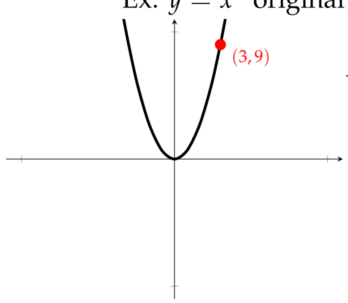


x	y
0	0
1	0.5
2	2
3	4.5
4	8
5	12.5
6	18

$y = f(x)$ vs. $y = -f(x)$

$y = f(x)$ original

Ex: $y = x^2$ original

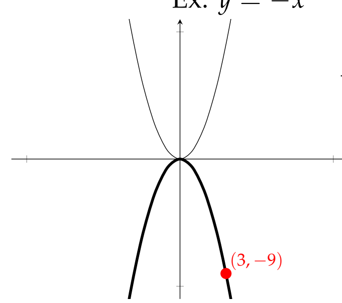


x	y
0	0
1	1
2	4
3	9
4	16
5	25
6	36

flip vert

$y = -f(x)$

Ex: $y = -x^2$



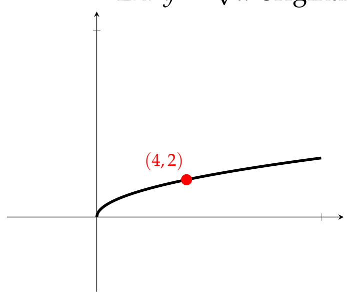
x	y
0	0
1	-1
2	-4
3	-9
4	-16
5	-25
6	-36

Horizontal transformations

Our example uses $f(x) = \sqrt{x}$. Because (a, b) is on the graph of $y = f(x)$ if and only if $f(a) = b$, we get...

$y = f(x)$
vs.
$y = f(x+c)$

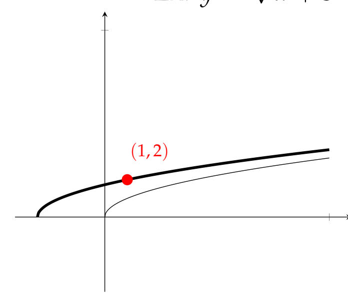
$y = f(x)$ original
Ex: $y = \sqrt{x}$ original



x	y
0	0
1	1
4	2
9	3
16	4
25	5
36	6

shift left by c
Example:
shift left by 3

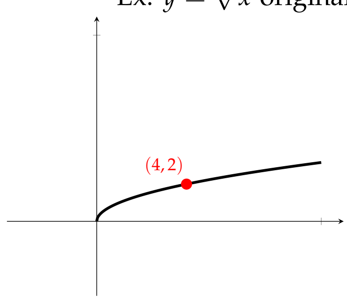
$y = f(x+c)$
Ex: $y = \sqrt{x+3}$



x	y
-3	0
-2	1
1	2
6	3
13	4
22	5
33	6

$y = f(x)$
vs.
$y = f(x-c)$

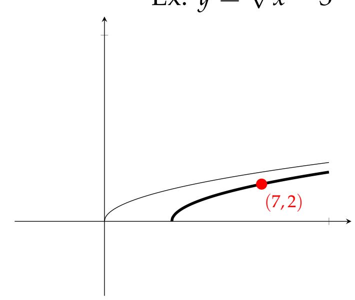
$y = f(x)$ original
Ex: $y = \sqrt{x}$ original



x	y
0	0
1	1
4	2
9	3
16	4
25	5
36	6

shift right by c
Example:
shift right by 3

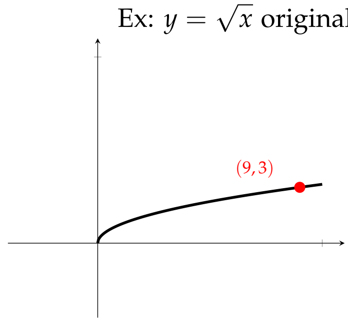
$y = f(x-c)$
Ex: $y = \sqrt{x-3}$



x	y
3	0
4	1
7	2
12	3
19	4
28	5
39	6

$y = f(x)$
vs.
$y = f(cx)$

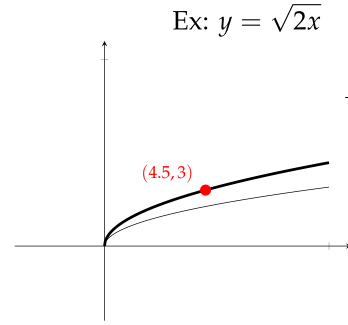
$y = f(x)$ original
Ex: $y = \sqrt{x}$ original



x	y
0	0
1	1
4	2
9	3
16	4
25	5
36	6

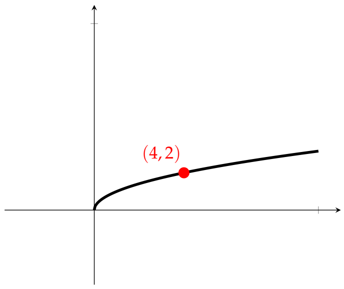
horiz shrink by c
Example:
horiz shrink by 2

$y = f(cx)$
Ex: $y = \sqrt{2x}$



x	y
0	0
0.5	1
2	2
4.5	3
8	4
12.5	5
18	6

<table>
<tr>
<td>$y = f(x)$
vs.
$y = f(\frac{1}{c}x)$</td>
<td>$y = f(x)$ original

Ex: $y = \sqrt{x}$ original

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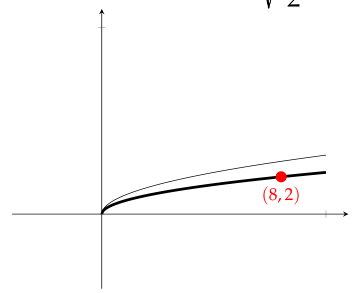
<tr><td>9</td><td>3</td></tr>
<tr><td>16</td><td>4</td></tr>
<tr><td>25</td><td>5</td></tr>
<tr><td>36</td><td>6</td></tr>
</table></td>
<td>horiz stretch by c
Example:
horiz stretch by 2</td>
<td>$y = f(\frac{1}{c}x)$
Ex: $y = \sqrt{\frac{1}{2}x}$

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<tr><th>x</th><th>y</th></tr>
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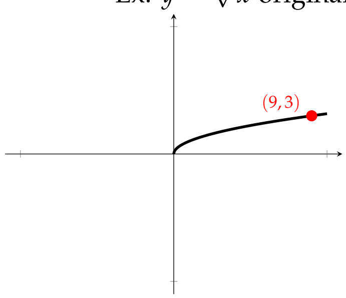
<tr><td>8</td><td>2</td></tr>
<tr><td>18</td><td>3</td></tr>
<tr><td>32</td><td>4</td></tr>
<tr><td>50</td><td>5</td></tr>
<tr><td>72</td><td>6</td></tr>
</table></td>
</tr>
</table>

<table>
<tr>
<td>$y = f(x)$
vs.
$y = f(-x)$</td>
<td>$y = f(x)$ original
Ex: $y = \sqrt{x}$ original

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<tr><th>x</th><th>y</th></tr>
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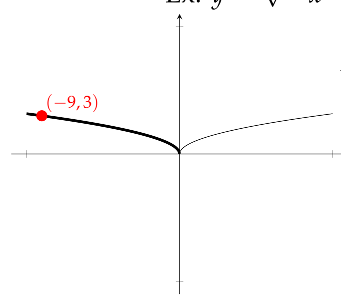
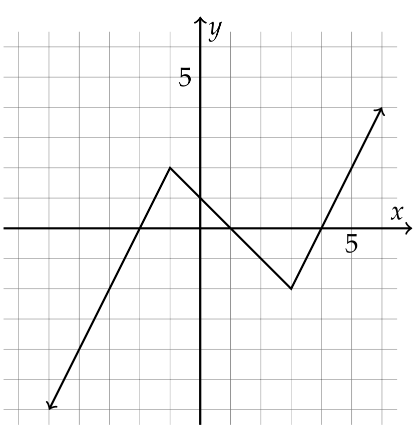
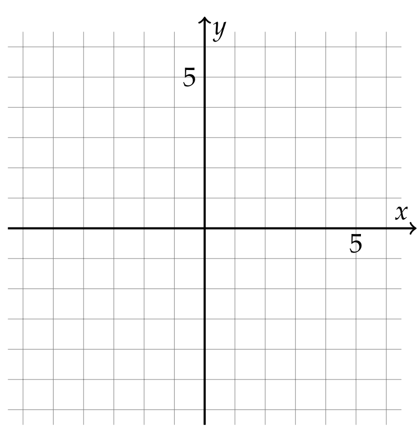
<tr><td>1</td><td>1</td></tr>
<tr><td>4</td><td>2</td></tr>
<tr><td>9</td><td>3</td></tr>
<tr><td>16</td><td>4</td></tr>
<tr><td>25</td><td>5</td></tr>
<tr><td>36</td><td>6</td></tr>
</table></td>
<td>flip horiz</td>
<td>$y = f(-x)$
Ex: $y = \sqrt{-x}$

<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>−1</td><td>1</td></tr>
<tr><td>−4</td><td>2</td></tr>
<tr><td>−9</td><td>3</td></tr>
<tr><td>−16</td><td>4</td></tr>
<tr><td>−25</td><td>5</td></tr>
<tr><td>−36</td><td>6</td></tr>
</table></td>
</tr>
</table>



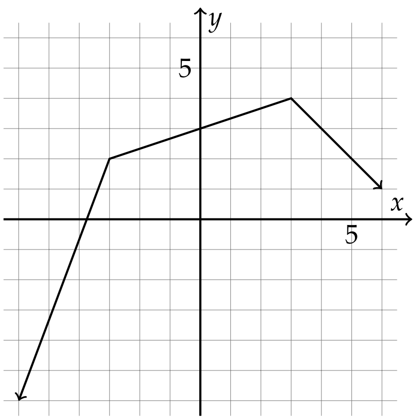
Graph of $y = f(x)$



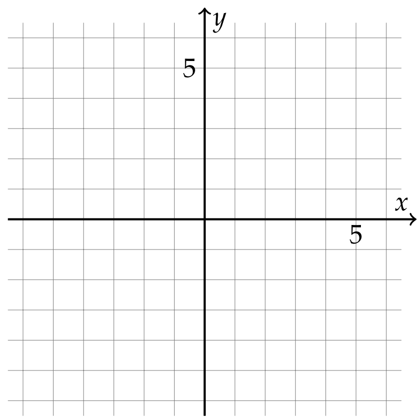
Graph of $y = f(x) + 3$

Based on the "original function" on the left, sketch the graph of the graph transformation

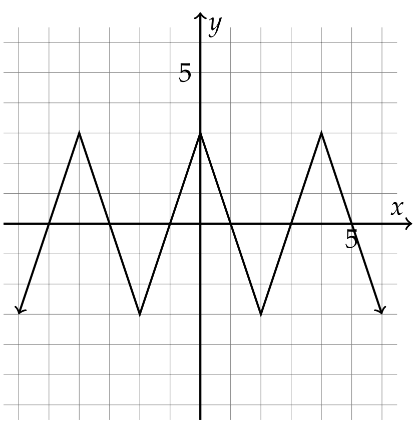
Carefully determine if the transformation is vertical or horizontal



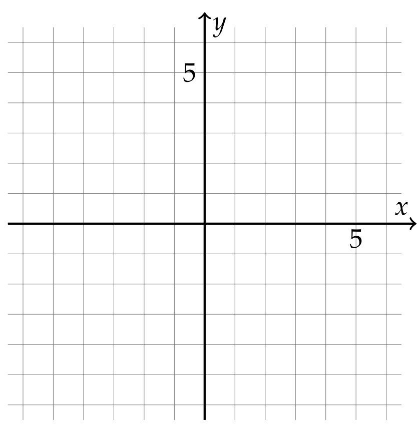
Graph of $y = g(x)$



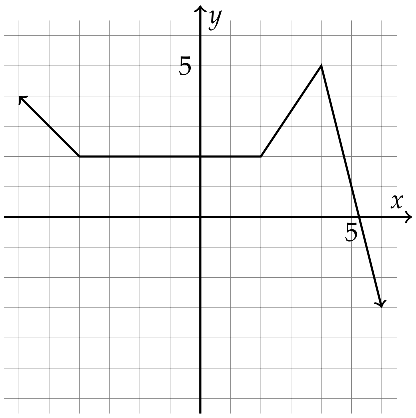
Graph of $y = g(x+3)$



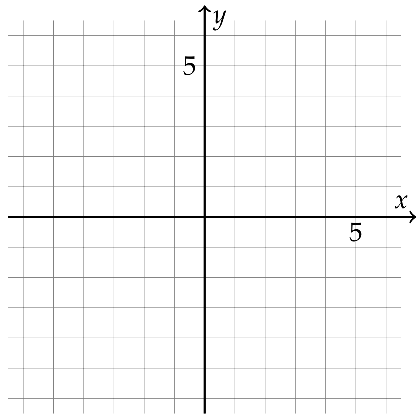
Graph of $y = h(x)$



Graph of $y = 2\,h(x)$

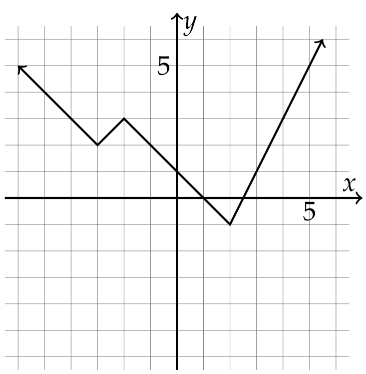


Graph of $y = k(x)$

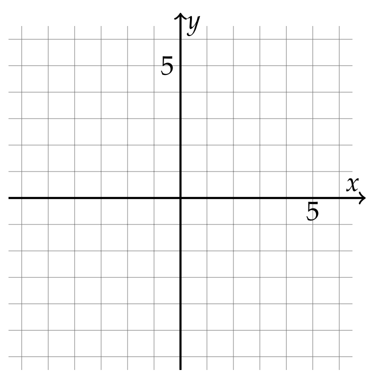


Graph of $y = k(2x)$

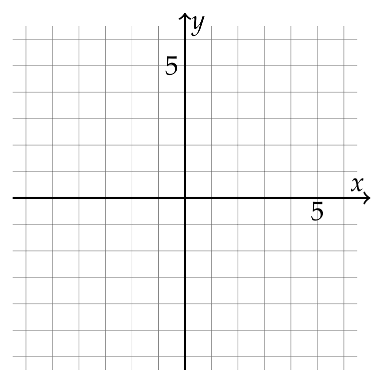
The suggested order:
1st, horiz stretch/shrink
2nd, horiz shift
3rd, vert stretch/shrink
4th, vert shift



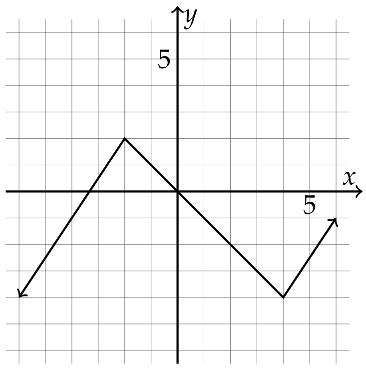
Graph of $y = f(x)$



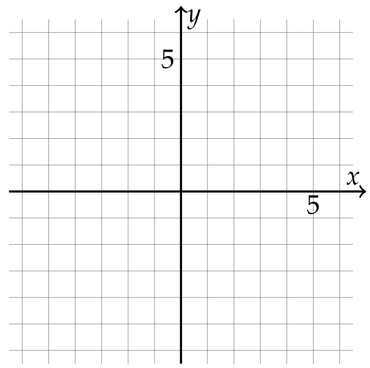
Graph of $y = f(x-2)$



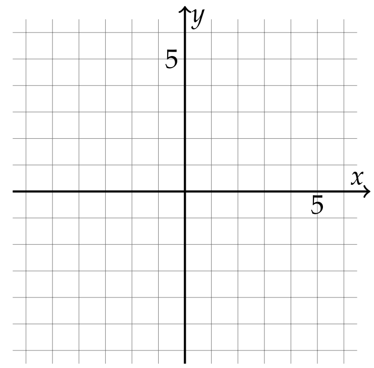
Graph of $y = f(x-2) - 3$



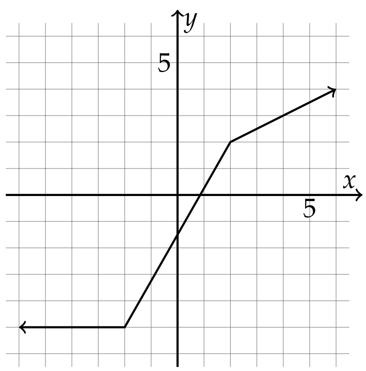
Graph of $y = g(x)$



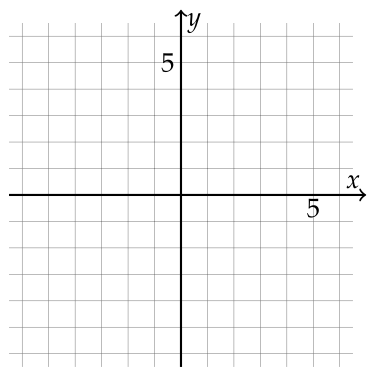
Graph of $y = g(2x)$



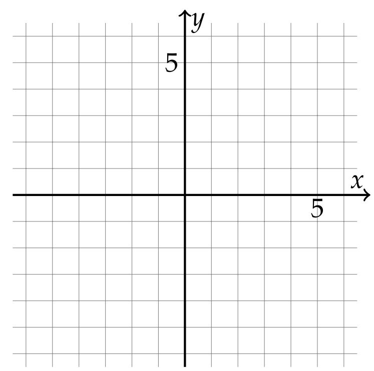
Graph of $y = g(2x) + 1$



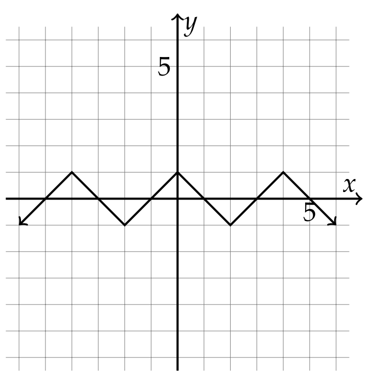
Graph of $y = h(x)$



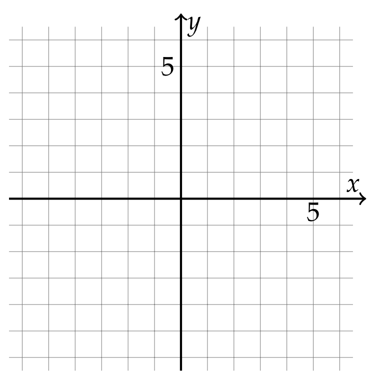
Graph of $y = h(x + 3)$



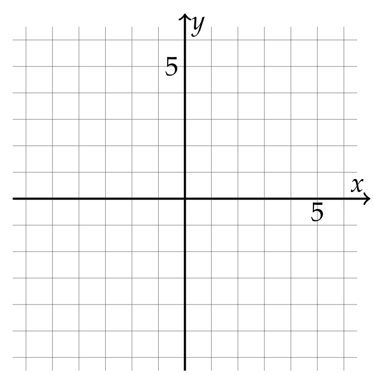
Graph of $y = -h(x + 3)$



Graph of $y = k(x)$



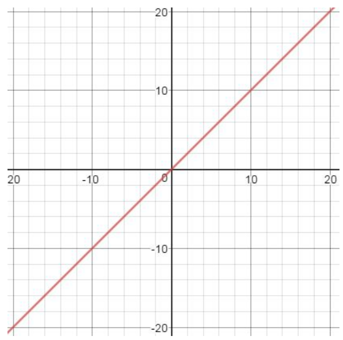
Graph of $y = 3\,k(x)$



Graph of $y = 3\,k(x) - 1$

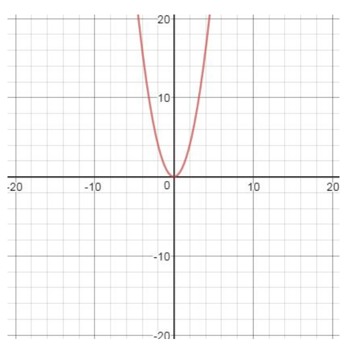
Base Graphs

$f(x) = x$



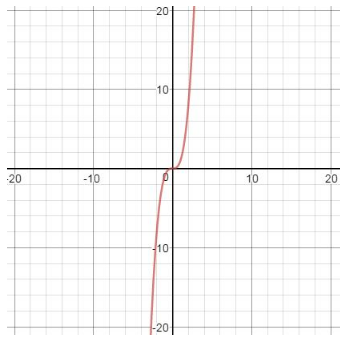
- Domain: $(-\infty, \infty)$
- Range: $(-\infty, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

$f(x) = x^2$



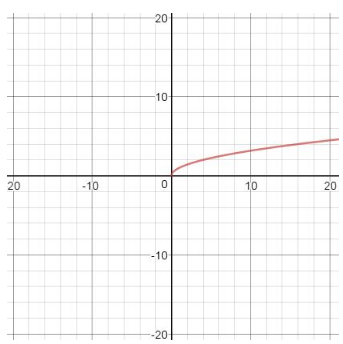
- Domain: $(-\infty, \infty)$
- Range: $[0, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

$f(x) = x^3$



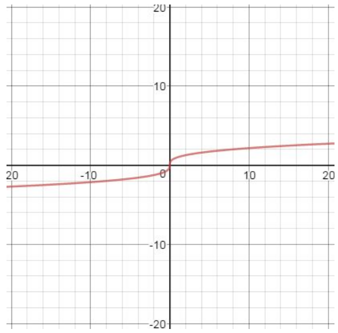
- Domain: $(-\infty, \infty)$
- Range: $(-\infty, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

$f(x) = \sqrt{x}$



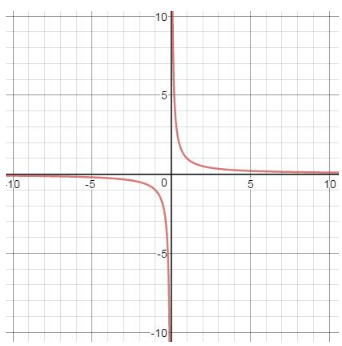
- Domain: $[0, \infty)$
- Range: $[0, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

$f(x) = \sqrt[3]{x}$



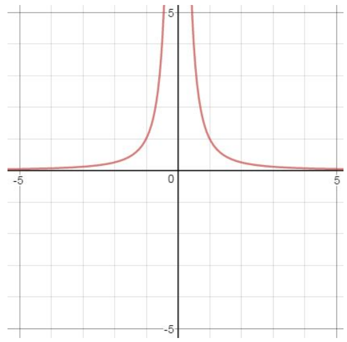
- Domain: $(-\infty, \infty)$
- Range: $(-\infty, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

$f(x) = \dfrac{1}{x}$



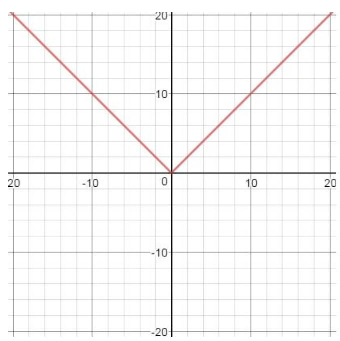
- Domain: $(-\infty, 0) \cup (0, \infty)$
- Range: $(-\infty, 0) \cup (0, \infty)$
- x-intercepts: None
- y-intercepts:None

$f(x) = \dfrac{1}{x^2}$



- Domain: $(-\infty, 0) \cup (0, \infty)$
- Range: $(0, \infty)$
- x-intercepts: None
- y-intercepts: None

$f(x) = |x|$



- Domain:$(-\infty, \infty)$
- Range: $[0, \infty)$
- x-intercepts: $(0, 0)$
- y-intercepts: $(0, 0)$

Transformations of Base Graphs

There are several types of graph transformations: Vertical shifts, horizontal shifts, reflections across the x-axis, reflections across the y-axis, vertical stretching, and horizontal stretching.

Vertical Shifts

Given a function $f(x)$, the graph of $f(x)+c$ ($c>0$) is the graph of $f(x)$ shifted *up* c units.

Given a function $f(x)$, the graph of $f(x)-c$ ($c>0$) is the graph of $f(x)$ shifted *down* c units.

Example 339. *Sketch the graph of* $g(x)=x^2+3$

Example 340. *Sketch the graph of* $h(x)=\sqrt{x}-2$

Horizontal Shifts

Given a function $f(x)$, the graph of $f(x+c)$ ($c>0$) is the graph of $f(x)$ shifted *left* c units.

Given a function $f(x)$, the graph of $f(x-c)$ ($c>0$) is the graph of $f(x)$ shifted *right* c units.

Example 341. *Sketch the graph of* $f(x)=\dfrac{1}{x+2}$

Example 342. *Sketch the graph of* $y=\dfrac{1}{(x-3)^2}$

Reflections

Given a function $f(x)$, the graph of $-f(x)$ is the graph of $f(x)$ reflected across the *x-axis*.

Given a function $f(x)$, the graph of $f(-x)$ is the graph of $f(x)$ reflected across the *y-axis*.

Example 343. *Sketch the graph of* $f(x)=|-x|$

Example 344. *Sketch the graph of* $f(x)=-|x|$

Vertical Stretching

Given a function $f(x)$, the graph of $cf(x)$ ($c>1$) is the graph of $f(x)$ stretched vertically by a factor of c.

Given a function $f(x)$, the graph of $cf(x)$ ($0<c<1$) is the graph of $f(x)$ compressed vertically by a factor of $\dfrac{1}{c}$.

Example 345. *Sketch the graph of* $f(x)=2x^2+4x+2$

Example 346. *Sketch the graph of* $g(x)=\dfrac{1}{2}(x-1)^2$

Horizontal Stretching

Given a function $f(x)$, the graph of $f(cx)$ $(c > 1)$ is the graph of $f(x)$ compressed horizontally by a factor of c.

Given a function $f(x)$, the graph of $f(cx)$ $(0 < c < 1)$ is the graph of $f(x)$ stretched horizontally by a factor of $\frac{1}{c}$.

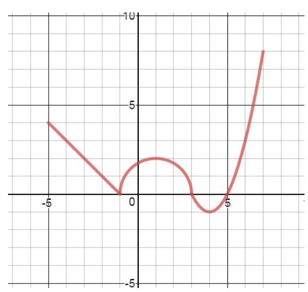
Example 347. *Sketch the graph of* $y = \sqrt[3]{8x}$

Example 348. *Sketch the graph of* $g(x) = \dfrac{1}{3x}$

Example 349. *Graph the function* $f(x) = -2(x-4) + 3$.

Example 350. *If* $P(2, -4)$ *is on the graph of* $f(x)$, *what is the corresponding point on the graph of* $g(x) = -f(2x) + 3$?

Example 351. *Below is the graph of* $f(x)$. *Graph* $g(x) = -f(x+1) + 3$.



Even and Odd Functions

A function is called **even** if

$$f(x) = f(-x).$$

Graphically, a function is even if it is symmetric about the y-axis.

A function is called **odd** if

$$-f(x) = f(-x).$$

Graphically, a function is even if it is symmetric about the origin.

Example 352. *Give two functions which are even. Then, sketch their graphs.Verify your two functions are even by showing that* $f(x) = f(-x)$.

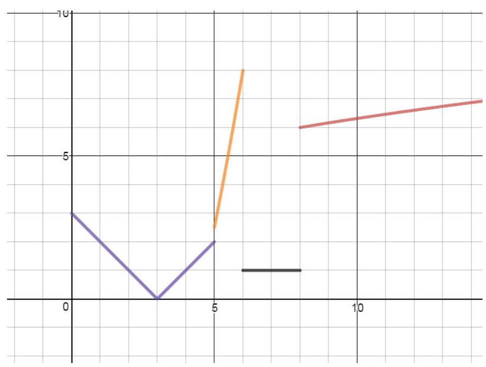
Example 353. *Give two functions which are odd. Then, sketch their graphs. Verify your two functions are odd by showing that* $-f(x) = f(-x)$.

Piece-wise Functions

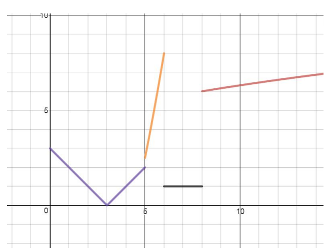
A **Piece-wise** defined function is a function which is defined in pieces. For example:

$$f(x) = \begin{cases} |x-3|, & 0 \le x < 5 \\ \frac{1}{2}x^2 - 10, & 5 \le x \le 6 \\ 1, & 6 < x \le 8 \\ \sqrt{x+1}+3 & , x > 8 \end{cases}$$

This looks like the following:



Example 354. *For the function,*



$$f(x) = \begin{cases} |x-3|, & 0 \le x < 5 \\ \frac{1}{2}x^2 - 10, & 5 < x \le 6 \\ 1, & 6 < x \le 8 \\ \sqrt{x+1}+3 & , x > 8 \end{cases}$$

- *What is the domain of this function?*
- *What is the range of this function?*
- *What is $f(0)$, $f(1)$, $f(5)$, $f(20)$?*
- *Sketch the graph of $g(x) = -f(-x)$. Then, find the rule for $g(x)$.*

Function operations

Formula 355

$(f+g)(x) = f(x) + g(x)$

Formula 356

$(f-g)(x) = f(x) - g(x)$

Formula 357

$(fg)(x) = f(x) \cdot g(x)$

Formula 358

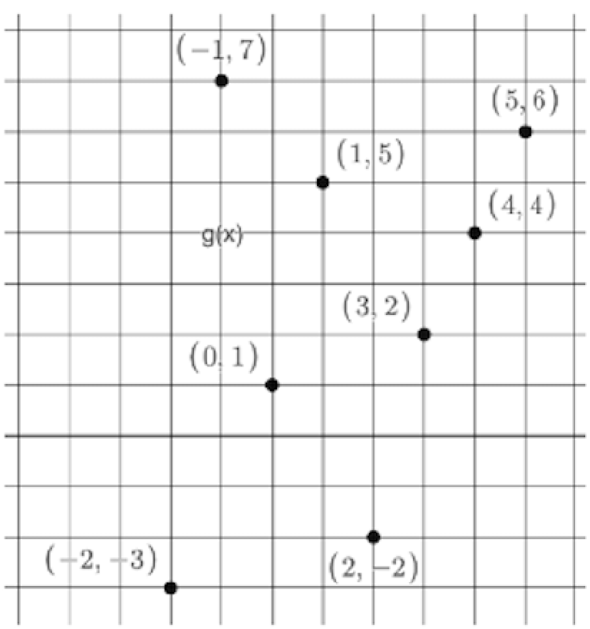
$(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$

Formula 359: Composition

$(f \circ g)(x) = f(g(x))$

Example 360. *If* $f(x) = (x+3)^x$ *and* $g(x) = 10(x^2 - 8x)$ *find* $(f \circ g)(x)$

Example 361. *If* $f(x) = (x+3)^x$ *and* $g(x) = 10(x^2 - 8x)$ *find* $(g \circ f)(x)$



x	$h(x)$
-3	5
-2	3
-1	1
0	-1
1	-3
2	-5
3	-7

The picture in the upper left shows the graph of a function called $g(x)$ while the table on the right shows the inputs/outputs for a function called $h(x)$. Let $f(x) = x^2 + 6$

Example 362. *Find* $(h \circ g)(3)$

Example 363. *Find* $(g+h)(2)$

Example 364. *Find* $(h \circ g \circ h)(-2)$

Example 365. *Find* $5 + g(5)$

Example 366. *Find* $10 + (f \circ h \circ g)(-2)$

One-to-one functions and Inverse Functions

1-1 Functions

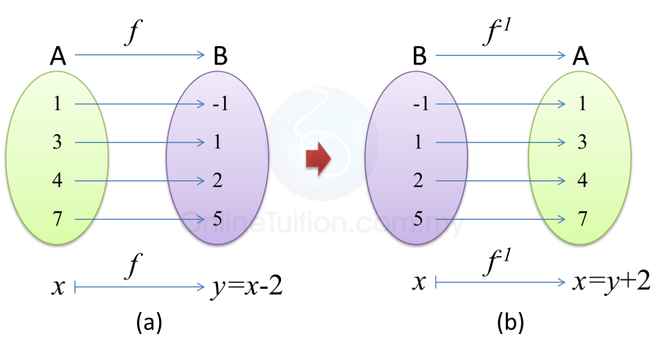
A function $f(x)$ is called $1-1$ (one-to-one) if for every output, there is exactly 1 corresponding input. Said another way, if $a \neq b$, then $f(a) \neq f(b)$. In other words, if $f(a) = f(b)$, then $a = b$. It turns out that $1-1$ functions pass the Horizontal Line Test (HLT): every horizontal line passes through the graph at most 1 time.

Inverse Functions

Given a $1-1$ function f, we define the inverse of f to be the function g so that

$$g(b) = a \text{ exactly when } f(a) = b.$$

The inverse of f is denoted f^{-1}.

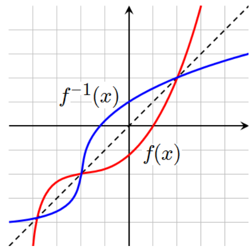


Properties of Inverses

Here is a list of properties about inverse functions:

- The domain of f is the range of f^{-1}.
- The range of f is the domain of f^{-1}.
- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$

- The graph of f^{-1} is the graph of f reflected across the line $y = x$



Example 367. *Show that* $f(x) = \dfrac{3}{x}$ *is its own inverse.*

Example 368. *Explain why* $f(x) = x^2$ *and* $g(x) = \sqrt{x}$ *are NOT inverse functions.*

Example 369. *Modify the domain of* $f(x) = x^2$ *so that* $f(x) = x^2$ *IS the inverse to* $g(x) = \sqrt{x}$.

Example 370. *If* $f(2) = 3$, $g^{-1}(3) = 5$, *and* $f(5) = 0$, *what is* $(f^{-1} \circ g^{-1} \circ f)(2)$?

Example 371. *Show that* $f(x) = 3x + 4$ *and* $f^{-1}(x) = \dfrac{x-4}{3}$ *are indeed inverses by showing that* $(f \circ f^{-1})(x) = x$ *and* $(f^{-1} \circ f)(x) = x$.

Example 372. *Graph both* $f(x) = 3x + 4$ *and* $f^{-1}(x) = \dfrac{x-4}{3}$ *to verify that they are reflections of each other across the line* $y = x$.

Method 373: How to Find the Inverse Function

Given a $1-1$ function $f(x)$

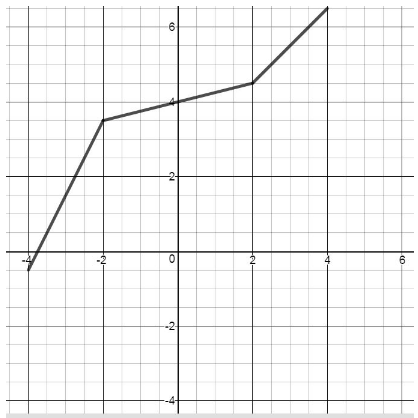
1. Change $f(x)$ to y.
2. Switch *ALL* x's and y's:
3. Solve for y:
4. Change y back to $f^{-1}(x)$

Example 374. *Find the inverse to* $f(x) = \dfrac{2x-4}{3+x}$ *and check your work.*

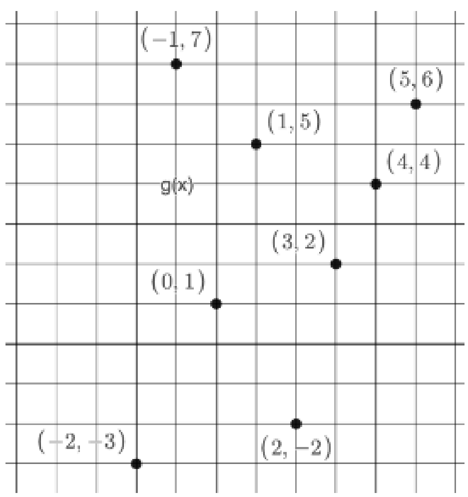
Example 375. *Find the inverse to* $f(x) = \dfrac{x+3}{5x-6}$ *and check your work.*

Example 376. *Find the inverse to* $f(x) = \sqrt[3]{x-2} + 4$ *and check your work.*

Example 377. *Given the graph of* f *below, sketch the graph of* f^{-1}.



6. Let $f(x) = \frac{2x+1}{3x-1}$, assume the graph below is the graph of $g(x)$ and note that g(x) is $1-1$ (why?). Finally, the table is the table of the $1-1$ function $h(x)$.



x	$h(x)$
-3	5
-2	3
-1	1
0	-1
1	-3
2	-5
3	-7

Compute the following:

A. $(g^{-1} \circ h^{-1})(3)$ B. $(h^{-1} \circ f \circ g^{-1})(1)$ C. $(h^{-1} fg)(1)$

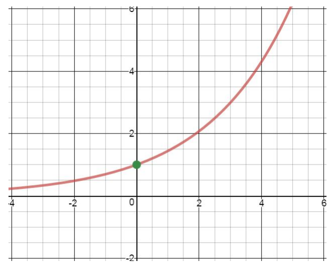
D. $(g^{-1} \circ h)(-3)$ E. $(h^{-1} \circ g \circ f^{-1})(-1)$ F. $(h \circ g \circ f)(1/8)$

G. $(f \circ h \circ g^{-1})(-2)$ H. $(h^{-1} \circ g^{-1} \circ g^{-1})(6)$ I. $f^{-1}\left(\frac{g^{-1}(2)}{g(4)}\right)$

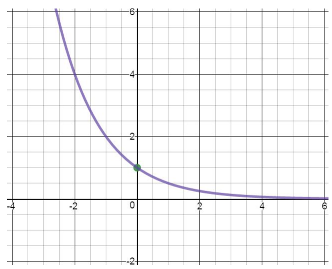
Exponential Functions

Exponential Functions

An exponential function is of the form $f(x) = a^x$, for a constant $a > 0$. If $a > 1$, the graph of $f(x)$ has the *general shape* given below:



If $0 < a < 1$, the graph of $f(x)$ has the *general shape* given below:



Properties of Exponential Functions

All exponential functions of the form $f(x) = a^x, a > 0$ pass through the point $(0, 1)$.

- $f(0) = a^0 = 1 \rightarrow (0, 1)$ is on the graph of f

Here are some additional properties:

- The domain of $f(x) = a^x$ is $(-\infty, \infty)$.
- The range of $f(x) = a^x$ is $(0, \infty)$.
- $f(x) = a^x$ has a horizontal asymptote at $y = 0$.
- $f(x) = a^x$ is a $1 - 1$ function and hence has an inverse.
- $a^x = a^y$ exactly when $x = y$.

The Number e

The number e is an irrational number between 2 and 3. In fact,

$$e \approx 2.71828182846$$

Since e is a number, it makes sense to talk about the exponential function $f(x) = e^x$.

Example 378. *Graph the function $f(x) = e^x$ and identify its domain and range.*

Example 379. *Graph the function* $f(x) = 2^{-x}$.

Example 380. *Graph the function* $g(x) = 3^{x+2}$.

Example 381. *Graph the function* $h(x) = -e^x - 4$.

Example 382. *Graph the exponential function of the form* $f(x) = Ba^{-x} + C$ *that has a horizontal asymptote at* $y = 72$, *y-intercept at 425, and passes through the point (1, 248.5).*

Example 383. *Solve* $\left(\frac{1}{2}\right)^{6-x} = 2$.

Example 384. *Solve* $9^{2x} \cdot \left(\frac{1}{3}\right)^{x+2} = 27 \cdot (3^x)^{-2}$.

Example 385. *Find the zeros to* $f(x) = x^2(2e^{2x}) + 4x(e^{2x}) + e^{2x}$.

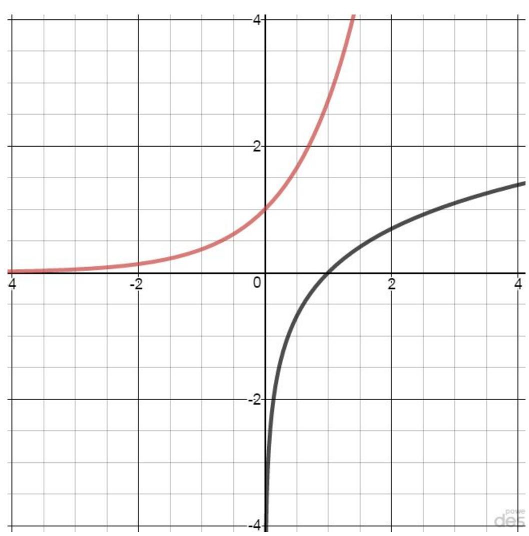
Example 386. *Given* $A = 1000$ *and* $r = 0.04$ *and* $t = 5$, *solve*[13] *for* P *in* $A = Pe^{rt}$.

[13] This is a **finance** problem, and the value of P answers the following question: if an account earns 4% interest compounded continuously, how much money should be deposited now so that the account balance is \$1000 in 5 years?

Logarithms

Defining the Logarithmic Functions

Recall that $f(x) = a^x$ is a $1-1$ function and hence has an inverse. In fact, we can even graph it:



If $f(x) = a^x$, then we know what $f^{-1}(x)$ *looks* like because it is reflected across the line $y = x$. We also know that the domain of $f^{-1}(x)$ is $(0, \infty)$ and that the range of $f^{-1}(x)$ is $(-\infty, \infty)$. Even better, we can *evaluate* $f^{-1}(x)$:

- If $f(x) = 2^x$, what is $f^{-1}\left(\frac{1}{4}\right)$?
- $f^{-1}\left(\frac{1}{4}\right) = b$ exactly when $f(b) = \frac{1}{4}$
- $f(b) = \frac{1}{4}$ means $2^b = 2^{-2} \to b = -2$
- So, $f^{-1}\left(\frac{1}{4}\right) = -2$.

We can ask a very natural question: What is this function? it turns out that given an exponential function $f(x) = a^x$, its inverse, $f^{-1}(x)$ is called the logarithm base a and is denoted:

$$f^{-1}(x) = \log_a(x).$$

Language Discussion 387: Logarithm

When seeing/writing $\log_a(x)$, say "log base a of x".
Do not leave out the word base and do no leave out the word of. The of is the same of in Language Discussion 287 to remind you we have a function (even though the name of the function is more than just the letter f).

The Logarithmic Function

For each $a > 0$, there is a different "log base a" function. And, there are two very special logarithms:

- $\log_{10}(x) = \log(x)$
- $\log_e(x) = \ln(x)$

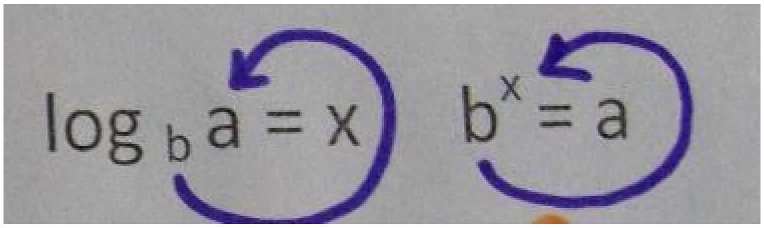
The domain for every log function is $(0, \infty)$. The range for every log function is $(-\infty, \infty)$. All log functions have the same shape and pass through $(1, 0)$. This means $\log_a(1) = 0$ for all $a > 0$.

Circle Trick

For every logarithmic equation, there is a corresponding exponential equation. This is useful because if you have an exponential equation you can't solve, convert it to a log equation. If you have a log equation you can't solve, convert it to an exponential equation. Recall that $f(x) = y \Leftrightarrow f^{-1}(y) = x$. For our two functions, this means

$$b^x = a \Leftrightarrow log_b(a) = x.$$

Method 388: Circle Trick



Example 389. *Evaluate the following logarithms:* $\log_9(3)$, $\log_2\left(\frac{1}{8}\right)$, $\log_4(-5)$.

Example 390. *Given* $A = 1000$ *and* $r = 0.04$ *and* $P = 850$, *solve*[14] *for* t *in* $A = Pe^{rt}$.

[14] This is a **finance** problem, and the value of t answers the following question: if an account earns 4% interest compounded continuously, how many years will it take an account with starting balance \$850 to eventually have a balance of \$1000?

Example 391. *Solve* $\log_2(x+9) = 4$.

Example 392. *Sketch the graph of* $\log_2|x| + 2$.

Logarithmic Properties

Below are a list of logarithmic properties

Formula 393

$\log_a(x) = \log_a(y)$ exactly when $x = y$.

Formula 394: Laws of Logarithms

- $\log_a(x^k) = k\log_a(x)$
- $\log_a(xy) = \log_a(x) + \log_a(y)$
- $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$

Example 395. *Use the log laws to expand the logarithms as much as possible:*

$$\ln\left(\frac{A^5B^3}{C^4D^2}\right)$$

.

Example 396. *Use the log laws to expand the logarithms as much as possible:*

$$\log\sqrt[3]{\frac{A}{B^2C^5}}$$

.

Example 397. *Use the log laws to expand the logarithms as much as possi-*

ble:

$$\log\left(\frac{\sqrt{x+6}}{x^4e^{2x}}\right)$$

.

Example 398. *Use the log laws to expand the logarithms as much as possible:*

$$\ln\sqrt{\frac{x+9}{3x+1}}$$

.

Example 399. *Use the log laws to expand the logarithms as much as possible:*

$$\log\left(\frac{x^2}{\sqrt[3]{x-2}(2x+3)^5}\right)$$

.

Example 400. *Rewrite* $\log_3(x)+\log_3(y)-7\log_3(z)$ *as one logarithm.*

Example 401. *Rewrite* $\ln(x)+\ln(y)-7\ln(z)$ *as one* natural *logarithm.*

Method 402

If $\log_a(x)=\log_a(y)$ then $x=y$.

Habit 403

If the original equation had variables in an input to any logarithm, check to ensure there are no false solutions. (Any solution which causes us to take the logarithm of 0 or negative is a false solution. This is because the input to any logarithm must be positive.)

Example 404. *Solve* $\ln(x+6)-\ln(10)=\ln(x-1)-\ln(2)$ *for* x.

Example 405. *Solve* $\log(x+4)=2-\log(x-2)$.

Formula 406: Logs and Exps are inverses

- $\log_a(a^x)=x$
- $a^{\log_a(x)}=x$

Example 407. *Simplify* $2^{10\log_2(m)}$

Example 408. *Simplify* $\log_3(9^x)$

Formula 409

$\log_a(1) = 0$

Formula 410: Change of base

$\log_a(x) = \dfrac{\ln(x)}{\ln(a)}$

Example 411. *Solve*[15] *for t in*

$$V = P \cdot \frac{1 - (1 + \frac{r}{m})^{-mt}}{\frac{r}{m}}$$

given $V = 215000$ *and* $r = 0.06$ *and* $m = 12$ *and* $P = 1100$. *State an exact answer, and then write what should be input into a calculator which only understands natural logarithms.*

[15] This is a **finance** problem. For a house loan of \$215000 with interest compounding at a rate of 6% monthly, and monthly payments of \$1100, finding t tells you how many years a family would have this mortgage.

Example 412. *Write*[16] *how should*

$$-\log_2(3.9 \times 10^{-4}) + \log_5 \frac{0.038}{0.183}$$

be input into a calculator

[16] This expression came from a chemistry problem.

Example 413. *Let* $f(x) = \ln(x - 2)$. *Use transformations to sketch the graph of* f. *Then find the domain and vertical asymptote. Let* $g(x) = \ln(x^2 + 2x - 15)$ *Use algebra (including a sign chart) to find the domain and vertical asymptotes. Then use desmos to sketch the graph of* g *and check your work*

Exponential and Logarithmic Equations

Example 414. *Solve for x in the equation* $7^{2x+3} = 3^{x-2}$.

Example 415. *Solve for x in the equation* $\log \sqrt{x - 9} = 2$.

Example 416. *Solve for x in the equation*
$\log_6(x - 4) - \log_6(3x - 10) = \log_6(\dfrac{1}{x})$.

Example 417. *Solve for x in the equation* $3 \log_5(x) = 2 \log_5(4)$.

Example 418. *Solve for x in the equation*
$\log(5x + 1) = \log(100) + \log(2x - 3)$

Example 419. *Solve for x in the equation* $\log(x^3) = (\log x)^3$.

Example 420. *Solve* $6e^{2x} = 1800$.

Example 421. *Solve* $2^t = 5^{7-4t}$.

Example 422. *Solve* $e^{2z} + 3e^z - 40 = 0$.

Example 423. *Solve* $\dfrac{2^x - 2^{-x}}{4} = 3$.

Formula 424: Exponential Growth/Decay

$$n(t) = n_o e^{rt},$$

where $n(t)$ is the population at time t, n_o is the initial size of population, r is the relative rate of growth, t is time.

Formula 425: Newton's Law of Cooling

$$T(t) = T_s + D_o e^{-kt},$$

where D_o is the initial temperature difference between object and surroundings, T_s is the temperature of the surroundings, t is time, and k is the constant dependent on object.

Example 426. *A species of bird was introduced to a certain island 25 years ago. Biologists observe that the population doubles every 10 years and now the population is 13,000. What was the initial population? What will the population be in 5 years? Graph the function that models this growth.*

Example 427. *A certain bacteria culture has a relative growth rate of 12% per hour but in the presence of an antibiotic, the relative growth rate is reduced to 5% per hour. The initial number of bacteria in the culture is 22. Find the projected population after 24 hours if (a) no antibiotic is present, (b) antibiotics are present.*

Example 428. *The half-life of cesium-17 is 30 years. Suppose we have a 10-g sample. How much of the sample will be left in 80 years? How long will it take for the sample to decrease to 2-g?*

Example 429. *Newton's Law of Cooling is used in homicide investigations to determine the time of death. The normal body temperature is 98.6 F. Immediately following death, the body begins to cool. It has been determined experimentally that the constant in Newton's Law of Cooling is approximately* $k = 0.1947$*, assuming time is measured in hours. Find a function that models the temperature t hours after death. Suppose the surrounding environment is 60 F. If the temperature of the body is now 72 F, how long ago was death?*

Appendix A: Completing the Square

Complete the square to solve a typical quadratic equation

$$3x^2 + 5x + 7 = 6$$

Subtract 7 on both sides.

$$3x^2 + 5x = -1$$

Divide by 3 on both sides. Be sure to divide the *entire* left side by 3.

$$\frac{3x^2 + 5x}{3} = \frac{-1}{3}$$

As written, the two 3's on the left side above don't cancel because the 3 in the numerator is not a factor. Rewrite.

$$\frac{3x^2}{3} + \frac{5x}{3} = \frac{-1}{3}$$

The rewrite above used the formula $\frac{r}{t} + \frac{s}{t} = \frac{r+s}{t}$. Now use $\frac{mp}{np} = \frac{m}{n}$ to cancel 3's.

$$x^2 + \frac{5x}{3} = \frac{-1}{3}$$

Because $\frac{5x}{3} = \frac{5}{3} \cdot \frac{x}{1} = \frac{5}{3}x$ using the formula $\frac{q}{r}\frac{s}{t} = \frac{qs}{rt}$, replace $\frac{5x}{3}$ with $\frac{5}{3}x$.

$$x^2 + \frac{5}{3}x = \frac{-1}{3}$$

With coefficient $\frac{5}{3}$, the length of second stick is $\frac{5}{2 \cdot 3}$. By squaring, $(\frac{5}{2 \cdot 3})^2 = \frac{5^2}{(2 \cdot 3)^2} = \frac{5^2}{4 \cdot 3^2} = \frac{25}{36}$. Add $\frac{25}{36}$ to both sides.

$$x^2 + \frac{5}{3}x + \frac{25}{36} = \frac{25}{36} - \frac{1}{3}$$

Factor left side. On right side, take the fraction after the minus sign and multiply by 12 on top and bottom.

$$\left(x + \frac{5}{6}\right)^2 = \frac{25}{36} - \frac{12}{36}$$

Simplify the right side by using the fraction subtraction formula $\frac{m}{p} - \frac{n}{p} = \frac{m-n}{p}$.

$$\left(x + \frac{5}{6}\right)^2 = \frac{13}{36}$$

Square root both sides, and put $\pm$ on one side.

$$\sqrt{\left(x + \frac{5}{6}\right)^2} = \pm\sqrt{\frac{13}{36}}$$

Square root and square undo each other on the left side.

$$x + \frac{5}{6} = \pm\sqrt{\frac{13}{36}}$$

Use $\sqrt[n]{\frac{c}{d}} = \frac{\sqrt[n]{c}}{\sqrt[n]{d}}$ to rewrite everything past the $\pm$ on the right side.

$$x + \frac{5}{6} = \pm\frac{\sqrt{13}}{\sqrt{36}}$$

Simplify square root in denominator. Note that $\sqrt{36}$ is 6 because $(6)^2 = 36$.

$$x + \frac{5}{6} = \pm\frac{\sqrt{13}}{6}$$

Subtract $\frac{5}{6}$ on both sides.

$$x = -\frac{5}{6} \pm \frac{\sqrt{13}}{6}$$

Use the fraction addition/subtraction rule $\frac{m}{p} \pm \frac{n}{p} = \frac{m \pm n}{p}$.

$$x = \frac{-5 \pm \sqrt{13}}{6}$$

Complete the square to get the Quadratic Formula This is the same type of work done in the specific example shown starting on page 84

$$ax^2 + bx + c = 0$$

Subtract c on both sides.

$$ax^2 + bx = -c$$

Divide by a on both sides. Be sure to divide the *entire* left side by a.

$$\frac{ax^2 + bx}{a} = \frac{-c}{a}$$

As written, the two a's on the left side above don't cancel because the a in the numerator is not a factor. Rewrite.

$$\frac{ax^2}{a} + \frac{bx}{a} = \frac{-c}{a}$$

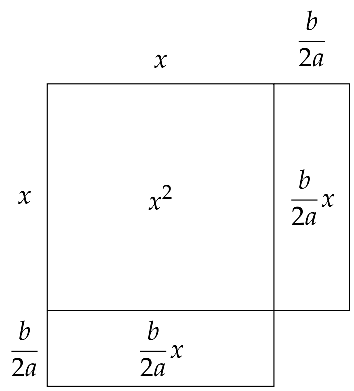
The rewrite above used the formula $\frac{r}{t} + \frac{s}{t} = \frac{r+s}{t}$.
Now use $\frac{mp}{np} = \frac{m}{n}$ to cancel a's.

$$x^2 + \frac{bx}{a} = \frac{-c}{a}$$

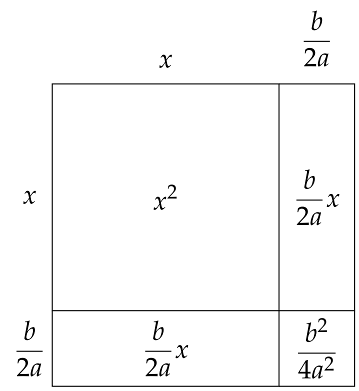
Because $\frac{bx}{a} = \frac{b}{a} \cdot \frac{x}{1} = \frac{b}{a}x$ using the formula $\frac{q}{r}\frac{s}{t} = \frac{qs}{rt}$, replace $\frac{bx}{a}$ with $\frac{b}{a}x$.

$$x^2 + \frac{b}{a}x = \frac{-c}{a}$$

- The linear coefficient is $\frac{b}{a}$, not $\frac{b}{a}x$. Do not include the x.
- Since the linear coefficient is $\frac{b}{a}$, the length of the second stick is $\frac{b}{a} \div 2 = \frac{b}{a} \div \frac{2}{1} = \frac{b}{a} \cdot \frac{1}{2} = \frac{b}{2a}$
- Since the second stick is $\frac{b}{2a}$, the constant to add to both sides is $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{(2a)^2} = \frac{b^2}{2^2a^2} = \frac{b^2}{4a^2}$



<u>Picture of</u> $x^2 + \frac{b}{a}x$



<u>Picture of</u> $x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}$

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}$$

Factor left side. On right side, take the fraction after the minus sign and multiply by $4a$ on top and bottom.

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{4ac}{4a^2}$$

Simplify the right side by using the fraction subtraction formula $\frac{m}{p} - \frac{n}{p} = \frac{m-n}{p}$.

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Square root both sides, and put $\pm$ sign on one side.

$$\sqrt{\left(x + \frac{b}{2a}\right)^2} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

Square root and square undo each other on the left side.

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

Use $\sqrt[n]{\dfrac{c}{d}} = \dfrac{\sqrt[n]{c}}{\sqrt[n]{d}}$ to rewrite everything past the $\pm$ on the right side.

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$$

Simplify square root in denominator. Note that $\sqrt{4a^2}$ is $2a$ because $(2a)^2 = 4a^2$.

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

Subtract $\dfrac{b}{2a}$ on both sides.

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Use the fraction addition/subtraction rule $\dfrac{m}{p} \pm \dfrac{n}{p} = \dfrac{m \pm n}{p}$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$